

PHYSICAL QUANTITIES & UNITS — MEMORY TRICK SHEET

General Science · Physics · Chapter 1 · Units + SI + Practical Units + Prefixes + Dimensions

Ye full notes nahi hai — source chapter ke scalar-vector pairs, unit systems, SI units, practical units, derived-unit confusions, prefixes aur dimensional formulae ko exam-recall tricks mein compress kiya gaya hai.

1. Physical Quantity — Measurable + Number

- Physical quantity = directly ya indirectly measurable property
- Fixed numerical value ke saath express ki jaati hai
- Examples: Mass, Length, Force

TRICK - “Measure + Number = Physical Quantity”

Naam ya concept tab physical quantity banta hai jab measurement ke context mein express ho.

2. Scalar vs Vector — M vs M+D

Scalar = Magnitude Only	Vector = Magnitude + Direction
Distance	Displacement
Speed	Velocity
Mass	Force
Work	Acceleration
Time	Momentum
Energy	—
Electric current	—
Temperature	—
Pressure	—

TRICK - “Scalar = Sirf Size; Vector = Size + Side”

Magnitude only = Scalar.

Magnitude + direction = Vector.

EXAM TRAP - “D-S-M / D-V-F”

Distance-Speed-Mass = Scalar.

Displacement-Velocity-Force = Vector.

3. Fundamental - Derived - Supplementary

Type	Dependency	Examples
Fundamental	Independent	Length, Mass, Time
Derived	Fundamental quantities par dependent	Force, Work, Pressure
Supplementary	Source convention: 2 quantities	Plane angle, Solid angle

TRICK - “F = Free, D = Dependent, S = 2 Angles”

Fundamental independent; Derived depends; Supplementary source convention mein Plane + Solid angle.

EXAM TRAP -

Source notes convention = 7 fundamental + 2 supplementary SI units.

4. Unit Systems — F-C-M-S

System	Identity	L-M-T	Source Key Fact
FPS	British	Foot-Pound-Second	1 Pound = 453 gm; source says India 1826

CGS	French / Metric	cm-g-s	Source says India implementation 1956
MKS	—	m-kg-s	Developed in France
SI	International	m-kg-s	Source links 1960 Paris conference; 7+2

TRICK - “FPS = Foot Pound Second; CGS = Centi Gram Second; MKS = Meter Kilo Second”

System ka naam hi L-M-T base set reveal karta hai.

TRICK - “SI = 1960 Paris + 7+2”

Source chapter ka historical recall pair.

5. Seven Fundamental SI Units — LMT-TELA

Quantity	Unit	Symbol
Length	Meter	m
Mass	Kilogram	kg
Time	Second	s
Temperature	Kelvin	K
Electric Current	Ampere	A
Luminous Intensity	Candela	cd
Amount of Substance	Mole	mol

TRICK - “LMT-TELA”

Length - Mass - Time - Temperature - Electric current - Luminous intensity - Amount of substance.

TRICK - “m kg s / K A cd mol”

Unit-symbol chain ko 3 + 4 ke block mein bolo.

Supplementary Quantity	Unit	Symbol
Plane Angle	Radian	rad
Solid Angle	Steradian	sr

6. Astronomy Distance Trio — PC > LY > AU

Unit	Value	Use
Parsec	3.26 light years = 3.08×10^{16} m	Astronomical distance
Light Year	9.46×10^{15} m	Light travels in one year
AU	1.496×10^{11} m	Earth-Sun distance

TRICK - “P-L-A = Parsec > Light Year > AU”

Distance order: PC > LY > AU.

EXAM TRAP - “Parsec = 3.26 LIGHT YEARS, not AU”

Direct source trap.

7. Tiny / Atomic / FPS Units

Unit	Value / Equivalent	Hook
Angstrom	10^{-10} m	Atomic distance
amu	1.66×10^{-27} kg	Atomic mass
Slug	14.59 kg	FPS mass
Svedberg	10^{-13} s	Sedimentation rate

TRICK - “A-10, amu-27, Svedberg-13”Angstrom = 10^{-10} m.amu = 1.66×10^{-27} kg.Svedberg = 10^{-13} s.

8. Special Practical Units — D-C-B-C

Unit / Limit	Used For / Value
Dobson	Ozone layer thickness
Cusec	Flow of water
BTU	Heat
Barrel	158.98 litres; oil volume
Chandrasekhar Limit	1.4 solar masses; white dwarf maximum mass

TRICK - “Dobson Ozone - Cusec Water - BTU Heat - Barrel Oil”

Special units ko use ke saath pair karo.

EXAM TRAP - “Chandra = 1.4 Sun”

Chandrasekhar limit = 1.4 solar masses, source notes ke mutabik.

9. Derived Units — J-N-W-P

Quantity	Unit	Symbol
Work / Energy	Joule	J
Force	Newton	N
Power	Watt	W
Pressure	Pascal	Pa
Potential Difference	Volt	V
Resistance	Ohm	Omega
Capacitance	Farad	F
Magnetic Flux	Weber	Wb
Frequency	Hertz	Hz
Charge	Coulomb	C

TRICK - “Work J, Force N, Power W, Pressure Pa”

J-N-W-P = four core mechanical derived units.

TRICK - “V-O-F-Wb-Hz-C”

Volt - Ohm - Farad - Weber - Hertz - Coulomb.

EXAM TRAP - “Weber = Flux; Watt = Power”

Wb aur W ka symbol confusion avoid karo.

10. Derived Unit Specials — Impulse = Momentum

Quantity	Unit
Momentum	kg m/s
Impulse	N s
Surface Tension	N/m
Angular Velocity	rad/s
Illuminance	Lux
Luminous Flux	Lumen

TRICK - “Impulse = Force x Time = Ns = Momentum pattern”

Impulse aur momentum ka dimensional formula same = $[MLT^{-1}]$.

11. Prefix Ladder — Exa to Atto

Power	Prefix	Symbol	Power	Prefix	Symbol
10^{18}	Exa	E	10^{-1}	Deci	d
10^{15}	Peta	P	10^{-2}	Centi	c
10^{12}	Tera	T	10^{-3}	Milli	m
10^9	Giga	G	10^{-6}	Micro	mu
10^6	Mega	M	10^{-9}	Nano	n
10^3	Kilo	k	10^{-12}	Pico	p
10^2	Hecto	h	10^{-15}	Femto	f
10^1	Deca	da	10^{-18}	Atto	a

TRICK - “Large: E-P-T-G-M-k-h-da”

18-15-12-9-6-3-2-1.

TRICK - “Small: d-c-m-mu-n-p-f-a”

-1,-2,-3,-6,-9,-12,-15,-18.

TRICK - “Mega +6 / Micro -6; Giga +9 / Nano -9; Tera +12 / Pico -12”

Opposite-sign mirror pairs.

EXAM TRAP - “Kilo = k, Kelvin = K”

Case-sensitive symbol trap.

12. Dimensions — Core Ladder

Quantity	Dimensional Formula
Density	$[ML^{-3}]$
Velocity / Speed	$[LT^{-1}]$
Acceleration	$[LT^{-2}]$
Momentum / Impulse	$[MLT^{-1}]$
Force	$[MLT^{-2}]$
Pressure / Stress / Modulus	$[ML^{-1}T^{-2}]$
Work / Energy	$[ML^2T^{-2}]$
Power	$[ML^2T^{-3}]$
Surface Tension	$[MT^{-2}]$
Frequency	$[T^{-1}]$

TRICK - “v - a - F - W - P = T exponent -1,-2,-2,-2,-3”

Velocity LT^{-1} ; Acceleration LT^{-2} ; Force MLT^{-2} ; Work ML^2T^{-2} ; Power ML^2T^{-3} .

TRICK - “Momentum = Impulse = MLT^{-1} ”

Same formula pair.

EXAM TRAP - “Pressure Family = $P-S-M = ML^{-1}T^{-2}$ ”

Pressure + Stress + Modulus of Elasticity.

13. Unique Dimension Patterns

Quantity	Unique Pattern
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Frequency	$[T^{-1}]$ only
Surface Tension	$[MT^{-2}]$ - no L
Density	$[ML^{-3}]$ - no T
Specific Heat	$[L^2T^{-2}K^{-1}]$
Resistance	$[ML^2T^{-3}A^{-2}]$
Capacitance	$[M^{-1}L^{-2}T^4A^2]$
Electric Field	$[MLT^{-3}A^{-1}]$

TRICK - “Freq only T; Surface no L; Density no T”

Fast elimination ke 3 strongest unique patterns.

14. Final Exam Traps — 10 Red Flags

- Speed scalar vs Velocity vector
- Distance scalar vs Displacement vector
- Mass scalar vs Force vector
- Source SI convention = 7 fundamental + 2 supplementary
- 1 Parsec = 3.26 light years, not AU
- Distance order = Parsec > Light Year > AU
- Magnetic flux = Weber (Wb), Power = Watt (W)
- Momentum = Impulse = $[MLT^{-1}]$
- Pressure = Stress = Modulus = $[ML^{-1}T^{-2}]$
- Kilo symbol k, Kelvin symbol K; Dobson-Ozone, Cusec-Water, BTU-Heat

EXAM TRAP - “S-V / D-D / M-F / 7+2 / P-L-A / Wb-W / M=I / PSM / k-K / DCB”

10 traps ko isi short chain mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Physical quantity = measurable + number → Scalar sirf magnitude, Vector magnitude+direction; D-S-M scalar vs D-V-F vector → Fundamental Free, Derived Dependent, Supplementary source 2 angles → systems FPS Foot-Pound-Second, CGS cm-g-s, MKS/SI m-kg-s; SI source 1960 Paris + 7+2 → LMT-TELA = m kg s K A cd mol; Plane rad, Solid sr → astronomy P-L-A = Parsec > Light Year > AU; Parsec = 3.26 LY → Angstrom 10^{-10} m, amu 1.66×10^{-27} kg, Slug 14.59kg, Svedberg 10^{-13} s → Dobson ozone, Cusec water, BTU heat, Barrel oil, Chandra 1.4Sun → derived J-N-W-P; V-Ohm-Farad-Weber-Hertz-Coulomb; Weber flux not Watt; Impulse Ns = Momentum kg m/s → prefix big E-P-T-G-M-k-h-da and small d-c-m-mu-n-p-f-a; Mega/Micro +/-6, Giga/Nano +/-9, Tera/Pico +/-12; k vs K → dimensions: Density ML^{-3} , v LT^{-1} , a LT^{-2} , Momentum/Impulse MLT^{-1} , Force MLT^{-2} , P-S-Modulus $ML^{-1}T^{-2}$, Work ML^2T^{-2} , Power ML^2T^{-3} , Surface MT^{-2} , Frequency T^{-1} .”

TOP 5: Scalar/Vector pairs | LMT-TELA | Parsec=3.26 LY + P>LY>AU | Weber vs Watt | Momentum=Impulse & Pressure=Stress=Modulus dimensions.

MOTION — MEMORY TRICK SHEET

General Science · Physics · Chapter 2 · Motion Types + Distance/Displacement + Graphs + Projectile + Circular Motion

Ye full notes nahi hai — source chapter ke 1D/2D/3D motion, distance-displacement, speed-velocity, equations, graph rules, projectile formulae aur circular-motion traps ko exam-recall tricks mein compress kiya gaya hai.

1. Motion Types — S-R-O

Type	Meaning	Examples
Straight Line	Straight path	Running child; straight-road car
Rotational	Own axis about circular path	Fan; bicycle wheel
Oscillatory	Back-and-forth about mean position	Pendulum; sewing-machine needle

TRICK - “S-R-O = Straight - Rotate - Oscillate”

Straight = line; Rotational = own axis; Oscillatory = mean position ke around back-forth.

2. Dimension of Motion — 1 Car, 2 Projectile, 3 Mosquito

Dimension	Coordinates	Examples
1D	x only	Car on straight road
2D	x + y	Projectile, circular motion, planet around Sun
3D	x + y + z	Mosquito, bird, water molecules

TRICK - “1-Car / 2-Projectile-Circle / 3-Mosquito”

1D straight road, 2D projectile/circular, 3D flying mosquito.

EXAM TRAP -

Projectile = 2D; Circular motion = 2D; Mosquito flying = 3D.

3. Uniform vs Non-Uniform — Equal vs Unequal

Motion	Condition	Distance-Time Graph
Uniform	Equal distances in equal time intervals	Straight line; constant slope
Non-Uniform	Unequal distances in equal time intervals	Curved line

TRICK - “Uniform = Equal-Equal = Constant Speed”

Distance-time graph ki constant slope speed ko show karti hai.

4. Distance vs Displacement — Path vs Shortcut

Feature	Distance	Displacement
Meaning	Actual path length	Shortest initial-to-final path
Quantity	Scalar	Vector
Value	Always positive	Positive, negative or zero
Formula	Length of path	Final position - Initial position

TRICK - “Distance = Drive Path; Displacement = Direct Shortcut”

Distance scalar; displacement vector.

EXAM TRAP - “Go 50 + Return 50 = Distance 100, Displacement 0”

Object move kare to distance zero nahi; displacement zero ho sakta hai.

5. Speed vs Velocity — d/t vs x/t

Feature	Speed	Velocity
Meaning	Distance/time	Displacement/time + direction
Quantity	Scalar	Vector
Value	Always positive	Positive, negative or zero
Average	Total distance / Total time	Total displacement / Total time

TRICK - “Speed = Scalar + Distance; Velocity = Vector + Displacement”

S-D / V-D pairing.

EXAM TRAP -

Speed scalar; Velocity vector. Same SI unit m/s, nature different.

6. Acceleration — Change in Velocity

- Acceleration = rate of change of velocity
- $a = \Delta v / \Delta t = (v-u)/t$
- Average acceleration = $(v_2-v_1)/(t_2-t_1)$
- Vector quantity; SI unit m/s^2
- Can be positive, negative or zero
- Negative acceleration = deceleration / retardation

TRICK - “Acceleration = Velocity Change / Time”

Negative a = deceleration/retardation.

EXAM TRAP -

Negative acceleration ko speed ke saath confuse mat karo; source terminology = deceleration/retardation.

7. Motion Symbols — U-V-A-T-S

Symbol	Meaning
u	Initial velocity
v	Final velocity
a	Acceleration
t	Time
s	Displacement

TRICK - “U-V-A-T-S”

Initial-Final-Acceleration-Time-Displacement.

8. Three Equations — V-S-V²

No.	Equation	Use
1	$v = u + at$	Final velocity after time t
2	$s = ut + \frac{1}{2}at^2$	Displacement after time t
3	$v^2 = u^2 + 2as$	Velocity-displacement relation

TRICK - “1st v, 2nd s, 3rd v-square”

Equation order ko variable se yaad karo: $v > s > v^2$.

TRICK - “Free Fall = a→g, s→h”

$v = u+gt$; $h = ut + \frac{1}{2}gt^2$; $v^2 = u^2 + 2gh$.

9. Graph Master Rule — Slope-Slope-Area

Graph	Slope Gives	Area Rule
Distance-Time	Speed	—
Displacement-Time	Velocity	—
Velocity-Time	Acceleration	Area = Displacement

TRICK - “s-t Slope = Speed/Velocity; v-t Slope = Acceleration; v-t Area = Displacement”

Graph questions ka master three-rule.

10. s-t Graph Shapes — Rest / Uniform / Accelerate / Decelerate

Shape	Motion
Horizontal line	Rest; $v=0$
Straight positive slope	Uniform motion
Upward curve	Uniform accelerated motion
Downward curve	Uniform decelerated motion

EXAM TRAP - “s-t Horizontal = STOP”

Horizontal s-t line means REST, not uniform motion.

11. v-t Graph Shapes — Constant / +Slope / -Slope

Shape	Motion
Horizontal line	Uniform motion; v constant; $a=0$
Straight + slope	Uniform acceleration
Straight - slope	Uniform deceleration
Upward curve	Non-uniform acceleration

TRICK - “v-t Horizontal = GO Constant”

s-t horizontal = rest, but v-t horizontal = uniform motion.

EXAM TRAP - “s-t H = Rest; v-t H = Uniform”

Same horizontal line, different meaning.

12. Projectile Formulae — T-H-R

Quantity	Formula
Time of Flight T	$2u \sin\theta / g$
Maximum Height H	$u^2 \sin^2\theta / 2g$
Range R	$u^2 \sin 2\theta / g$

TRICK - “Projectile = T-H-R”

Time-Height-Range formula trio.

- Path = Parabola
- Horizontal velocity remains constant if air resistance neglected
- Vertical velocity changes due to gravity
- At highest point vertical velocity = 0

TRICK - “Top Point = Vertical 0, Horizontal Continues”

Highest point par only vertical component zero.

13. Projectile Angles — 45 Range, 90 Height

Condition	Source Result
$\theta = 45^\circ$	Maximum Range
$\theta = 90^\circ$	Maximum Height
At maximum range	$R_{\max} = 4 \times H_{\max}$ - as stated in source notes

TRICK - “45 = Range; 90 = Height”

45 ko height aur 90 ko range mat karna.

TRICK - “Projectile = Parabola”

Source ideal condition under no-air-resistance framing.

EXAM TRAP -

Source notes state: 45° maximum range, 90° maximum height, and $R_{\max} = 4 \times H_{\max}$.

14. Circular Motion — Constant Speed, Changing Velocity

- Uniform circular motion: speed constant, velocity variable because direction changes
- Non-uniform circular motion: speed and velocity both variable
- Examples: electron around nucleus; Moon around Earth

TRICK - “Circle = Speed Same, Direction Changes, So Velocity Changes”

Uniform circular motion accelerated motion hai because velocity direction changes.

EXAM TRAP -

Constant speed ka matlab zero acceleration nahi in uniform circular motion.

15. Circular Formula Pack — Theta-Omega-v-alpha-a

Quantity	Formula / Direction
Angular displacement θ	Arc Δs / radius r
Angular velocity ω	$\Delta\theta / \Delta t$
Linear-angular relation	$v = \omega r = 2\pi nr$
Angular acceleration α	$\alpha = \omega/t$ - source formula
Centripetal acceleration	$a = v^2/r = r\omega^2$; towards centre

TRICK - “ $\theta \rightarrow \omega \rightarrow v = \omega r \rightarrow a_c = v^2/r$ ”

Circular-motion formula chain.

EXAM TRAP - “Centripetal = Centre; Centrifugal = Outward Pseudo”

Direction trap.

16. Master Scalar-Vector Table

Quantity	Nature	Unit
Distance	Scalar	m
Displacement	Vector	m
Speed	Scalar	m/s
Velocity	Vector	m/s
Acceleration	Vector	m/s ²
Angular velocity	Vector	rad/s
Angular acceleration	Vector	rad/s ²

TRICK - “D-S Scalar; D-V-A- ω - α Vector”

Distance + Speed = scalar.

Displacement + Velocity + Acceleration + angular quantities = vector.

17. Final Exam Traps — 8 Red Flags

- Projectile + circular motion = 2D; straight-road car = 1D; mosquito = 3D
- Distance scalar and always positive; displacement vector and can be zero
- Speed scalar; velocity and acceleration vectors
- Negative acceleration = deceleration / retardation
- s-t slope = speed/velocity; v-t slope = acceleration; v-t area = displacement
- Horizontal s-t = rest; horizontal v-t = uniform motion
- Projectile: 45° maximum range; 90° maximum height; path parabola
- Uniform circular motion = constant speed but variable velocity, so acceleration non-zero

EXAM TRAP - “2D-3D / D-D / S-V-A / Negative-a / Slope-Area / Horizontal / 45-90 / Circle”

Chapter ke 8 major traps isi chain mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Motion S-R-O = Straight-Rotational-Oscillatory → dimensions 1-Car, 2-Projectile/Circle, 3-Mosquito → Uniform equal-equal constant speed, Nonuniform unequal → Distance path-scalar-positive vs Displacement shortcut-vector +/-0; Speed distance/time scalar vs Velocity displacement/time vector → Acceleration $\Delta v/\Delta t$, negative = deceleration; UVATS symbols → equations 1st $v=u+at$, 2nd $s=ut+\frac{1}{2}at^2$, 3rd $v^2=u^2+2as$; free fall $a=g, s \rightarrow h$ → graphs: s-t slope speed/velocity, v-t slope acceleration, v-t area displacement; s-t horizontal REST, v-t horizontal UNIFORM → Projectile THR = $2u\sin\theta/g$, $u^2\sin^2\theta/2g$, $u^2\sin 2\theta/g$; path parabola; top vertical $v=0$; 45 range, 90 height; source $R_{\max}=4xH_{\max}$ → Circular: constant speed but variable velocity; $\theta=\text{arc}/r$, $\omega=\Delta\theta/\Delta t$, $v=\omega r=2\pi nr$, source $\alpha=\omega/t$, centripetal $a=v^2/r=r\omega^2$ toward centre; centrifugal outward pseudo.”

TOP 5: Distance vs Displacement | Speed vs Velocity | s-t/v-t slope-area rules | 45° Range vs 90° Height | Uniform circular motion = constant speed but non-zero acceleration.

LAWS OF FORCE & MOTION — MEMORY TRICK SHEET

General Science · Physics · Chapter 3 · Force + Newton Laws + Momentum + Friction + Circular Force + Torque + Lever

Ye full notes nahi hai — source chapter ke force-types, Newton's laws, momentum-impulse, friction order, centripetal/centrifugal force, torque-couple aur lever classes ko fast exam-recall tricks mein compress kiya gaya hai.

1. Force — Push/Pull + Shape/Direction/Motion

- Force object ki shape, direction ya state of motion ko change kar sakta hai
- Daily-life form = push, pull or hit
- SI unit = Newton (N); CGS unit = Dyne
- $1 \text{ N} = 10^5 \text{ dyne}$; $1 \text{ dyne} = 10^{-5} \text{ N}$

TRICK - "Force = Shape - Side - State"

Shape, direction aur state of motion ko change karne wala external factor.

TRICK - "N to Dyne = $\times 10^5$ "

1 Newton = 100000 dyne.

2. Force Types — C/N + B/U

Type	Meaning	Examples / Result
Contact	Physical contact required	Frictional, muscular
Non-Contact	No physical contact	Gravitational, magnetic
Balanced	Resultant = 0	$a = 0$; rest OR constant velocity
Unbalanced	Resultant $\neq 0$	$a \neq 0$; motion state changes

TRICK - "C-N = Contact-No Contact; B-U = Balance-Unbalance"

Balanced = net force zero.

Unbalanced = net force non-zero.

TRICK - "200 opposite 200 = 0; 400 opposite 200 = 200 toward larger"

Source example se resultant-force arithmetic lock karo.

EXAM TRAP -

Balanced force ka matlab hamesha rest nahi; constant velocity bhi ho sakti hai.

3. Newton's First Law — Inertia

- Object at rest remains at rest; moving object uniform straight-line motion continue karta hai unless external force acts
- Also called Law of Inertia
- Associated with Galileo's idea of inertia
- Inertia = change oppose karne ki tendency
- Inertia depends on mass; greater mass = greater inertia
- Types = inertia of rest, motion and direction

TRICK - "1st = Inertia; Mass = Measure of Inertia"

First law qualitative idea of force deta hai.

TRICK - "R-M-D = Rest - Motion - Direction"

Inertia ke 3 types.

EXAM TRAP -

$F = ma$ First Law se nahi; Second Law se milta hai.

4. Momentum + Newton's Second Law

- Linear momentum $p = m \times v$
- Momentum vector quantity hai
- SI unit = $\text{kg}\cdot\text{m/s}$
- Newton's Second Law: rate of change of momentum applied force ke proportional
- For constant mass: $F = ma$

TRICK - " $p = mv$; $2^{\text{nd}} = dp/dt = ma$ "

Momentum formula aur quantitative force law ek chain.

EXAM TRAP - "1 Inertia - 2 $F=ma$ - 3 Action/Reaction"

Newton laws ka master code.

5. Impulse — Force-Time = Momentum Change

- Impulse $I = F \times \Delta t$
- $I = \Delta p = m(v-u)$
- SI unit = $\text{N}\cdot\text{s}$
- Examples: hammer strike; catch lete waqt player hands backward move karta hai

TRICK - " $I = F\Delta t = \Delta p$ "

Impulse = force-time product = change in momentum.

TRICK - " $10\text{ g}, 0 \rightarrow 10\text{ m/s} = 0.1\text{ N}\cdot\text{s}$ "

Source worked example: $0.01 \times 10 = 0.1\text{ N}\cdot\text{s}$.

6. Newton's Third Law — Action-Reaction

- Every action has equal and opposite reaction
- $F_1 = -F_2$
- Equal magnitude, opposite direction
- Action-reaction forces different bodies par act karte hain
- Examples: rocket launch, rowing, swimming, bouncing ball, gun recoil

TRICK - "3rd = Equal-Opposite-Different Bodies"

Different bodies sabse important trap.

EXAM TRAP -

Action-reaction forces cancel nahi karte because they act on different bodies.

7. Conservation of Linear Momentum

- Net external force = 0 ho to total linear momentum constant
- $F = dp/dt = 0 \Rightarrow p = \text{constant}$
- $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$
- Examples: rocket propulsion, gun recoil, person jumps from boat

TRICK - "External Force ZERO $\rightarrow$ Total Momentum SAME"

Momentum conservation ki necessary source condition.

TRICK - " $10 \times 20 + 4 \times (-20) = 14v \rightarrow v = 8.57\text{ m/s right}$ "

Source collision example ka final answer.

EXAM TRAP -

Rocket motion source mein momentum conservation + Newton's Third Law dono se linked hai.

8. Friction — Formula + Cause

- Friction contact surfaces ke along tangentially act karta hai
- Relative motion ko oppose karta hai
- Cause: surface irregularities + molecular bonding at contact points
- $F = \mu R$

TRICK - "Friction = $\mu \times$ Normal Reaction"

$F = \mu R$; μ = coefficient of friction.

9. Friction Types — L-S-S-R

Type	Meaning / Condition
Static	Before motion; self-adjusting
Limiting	Maximum static; body about to move; $F_s = \mu_s R$
Kinetic	Moving contact surfaces
Sliding	Type of kinetic friction
Rolling	Least among listed friction types

TRICK - "Limiting > Static > Sliding > Rolling"

Source friction order.

TRICK - "Rolling = Least → Wheels/Ball Bearings"

Friction reduction ka reason.

EXAM TRAP -

Limiting friction = maximum static friction.

10. Friction — Uses, Losses, Control

Increase Friction	Decrease Friction
Grooves in tyres	Polish rough surfaces
Sand on snow roads	Lubricants
Rough shoe soles	Low-friction materials
—	Ball bearings
—	Streamlined shapes in fluids

TRICK - "Grip = Grooves-Sand-Rough; Glide = Polish-Lube-Bearing-Streamline"

Increase vs decrease friction quick split.

- Advantages: walking, brakes, writing on blackboard
- Disadvantages: energy loss, lower machine efficiency, wear & tear

11. Centripetal vs Centrifugal

Feature	Centripetal	Centrifugal
Direction	Towards centre	Away from centre
Nature	Real/resultant inward force	Apparent / pseudo in rotating frame
Magnitude form	$F = mv^2/r$	Same magnitude form; opposite apparent direction
Examples	Car circular path; electron around nucleus	Cream separator; washing-machine dryer

TRICK - “PETAL Pulls IN; FUGAL Flies OUT”

Centripetal inward real; centrifugal outward pseudo.

EXAM TRAP -

Centripetal = centreward real; Centrifugal = outward apparent/pseudo.

12. Circular Road Formulae

Situation	Formula
Maximum speed on level circular road	$v_{\max} = \sqrt{\mu_s r g}$
Banking relation	$\tan \theta = v^2/(r g)$

TRICK - “Level Road = $\sqrt{\mu r g}$; Bank = $\tan \theta = v^2/(r g)$ ”

Circular-road formula pair.

13. Torque + Couple

- Torque = force ka turning tendency about fixed axis
- $\tau = r \times F = r F \sin \theta$
- Vector quantity
- SI unit = $\text{N}\cdot\text{m}$
- Couple = two equal and opposite parallel forces with separated lines of action
- Moment of couple = Force $\times$ perpendicular separation

TRICK - “Torque = $r F \sin \theta$; Unit $\text{N}\cdot\text{m}$ ”

Turning effect formula + unit.

TRICK - “Couple = Equal-Opposite-Parallel + Gap”

Gap = perpendicular distance between force lines.

EXAM TRAP -

Torque unit $\text{N}\cdot\text{m}$ hai; source chapter mein vector quantity bhi stated hai.

14. Lever Basics — P-F-W + Arms

Term	Meaning
Effort P	Applied force
Fulcrum F	Rotation point
Load W	Load to be lifted
Effort arm a	Effort-to-fulcrum distance
Load arm b	Load-to-fulcrum distance

TRICK - “P-F-W / a-b”

P effort, F fulcrum, W load; a effort arm, b load arm.

- Principle of moments: $P \times a = W \times b$
- Mechanical Advantage $MA = W/P = a/b$

TRICK - “ $Pa = Wb$; $MA = W/P = a/b$ ”
Lever ka two-formula pack.

15. Lever Classes — Middle Person Rule

Class	Middle	MA	Examples
1st	Fulcrum	Can be >1 , $=1$ or <1	Scissors, see-saw, hammer, pliers, beam balance
2nd	Load	Always >1	Wheelbarrow, lemon squeezer, nutcracker, wheel & axle
3rd	Effort	Always <1	Tongs, ladder, plough

TRICK - “1-F, 2-L, 3-E”

Class number se middle element yaad karo.

TRICK - “L-F-E / F-L-E / F-E-L”

1st = Fulcrum middle; 2nd = Load middle; 3rd = Effort middle.

EXAM TRAP - “2nd MA >1 ; 3rd MA <1 ; 1st Any”

Mechanical advantage trap.

16. Final Exam Traps — 10 Red Flags

- Balanced force $\rightarrow a=0$; object constant velocity se move bhi kar sakta hai
- Gravitational + magnetic = non-contact
- 1st Inertia; 2nd $F=ma$; 3rd Action-Reaction
- Action-reaction different bodies par act karte hain; cancel nahi karte
- Impulse = $F\Delta t = \Delta p$; unit $N\cdot s$
- Momentum conserved only when net external force = 0
- Limiting = max static; Rolling = least friction
- Centripetal inward real; Centrifugal outward pseudo
- Torque $\tau = rF \sin\theta$; unit $N\cdot m$
- Lever: 1st F middle; 2nd L middle; 3rd E middle

EXAM TRAP - “B-N / 1-2-3 / I / M / Friction / Circular / Torque / Lever”

Chapter ke 10 red flags ko isi sequence mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Force shape-side-state; SI N, CGS dyne, $1N=10^5$ dyne $\rightarrow$ types Contact/Noncontact + Balanced/Unbalanced; balanced $a=0$ but constant- v possible $\rightarrow$ Newton 1 Inertia, mass measure; inertia R-M-D $\rightarrow$ momentum $p=mv$; 2nd $dp/dt \rightarrow F=ma$; impulse $I=F\Delta t=\Delta p$ $N\cdot s$ $\rightarrow$ 3rd equal-opposite different bodies; rocket-row-swim-recoil $\rightarrow$ momentum conservation external $F=0$, $m_1u_1+m_2u_2=m_1v_1+m_2v_2$ $\rightarrow$ friction $F=\mu R$; order Limiting>Static>Sliding>Rolling; rolling least; grip by grooves/sand/rough, reduce by polish/lube/bearings/streamline $\rightarrow$ centripetal IN real $F=mv^2/r$ vs centrifugal OUT pseudo; level-road $v_{max}=\sqrt{\mu sr}$, banking $\tan\theta=v^2/rg$ $\rightarrow$ torque $\tau=rF\sin\theta$ $N\cdot m$; couple equal-opposite-parallel with gap $\rightarrow$ lever $Pa=Wb$, $MA=W/P=a/b$; classes 1-F,2-L,3-E; 2nd $MA>1$, 3rd <1 , 1st any.”

TOP 5: Balanced force $\neq$ always rest | 1-2-3 Laws | Impulse= Δp | Limiting>Static>Sliding>Rolling | Lever 1-F, 2-L, 3-E.

WORK, POWER & ENERGY — MEMORY TRICK SHEET

General Science · Physics · Chapter 4 · Work + Power + KE/PE + Conservation + Energy Conversion

Ye full notes nahi hai — source chapter ke work-sign rules, conservative/non-conservative force, power conversions, energy units, KE/PE formulae, free-fall conservation aur device-wise energy conversions ko fast exam recall format mein compress kiya gaya hai.

1. Work — $F \cdot s \cdot \cos\theta$

- Work tab hota hai jab applied force displacement produce kare
- $W = F \cdot s \cdot \cos\theta$
- Work scalar quantity hai; sign angle θ par depend karta hai
- SI / MKS unit = Joule (J); CGS unit = erg
- $1 \text{ J} = 10^7 \text{ erg}$; $1 \text{ erg} = 1 \text{ dyne} \times 1 \text{ cm}$

TRICK - “Work = Force × Shift × $\cos\theta$ ”

F-s-cos θ ko formula order mein yaad karo.

cos θ hi positive/negative/zero sign decide karta hai.

TRICK - “J to erg = $\times 10^7$ ”

$1 \text{ J} = 10^7 \text{ erg}$; reverse = $1 \text{ erg} = 10^{-7} \text{ J}$.

2. Work Sign — + / 0 / -

Work	Angle	Source Example
Positive	$0^\circ \leq \theta < 90^\circ$	Gravity on freely falling body
Zero	$\theta = 90^\circ$ or 270°	Lifting load sideways; circular motion
Negative	$90^\circ < \theta < 270^\circ$	Gravity while lifting body upward

TRICK - “Same Side + ; Right Angle 0 ; Opposite -”

Falling body: gravity and displacement same direction → + work.

Lifting body: gravity opposite displacement → - work.

EXAM TRAP - “Circular = 90° = Work ZERO”

Source trap: force perpendicular to displacement in circular motion.

3. Conservative vs Non-Conservative

Feature	Conservative	Non-Conservative
Work depends on	Initial + final positions only	Initial/final + path
Path	Independent	Dependent
Closed path work	= 0	$\neq 0$
Source examples	Gravitation, electrostatic, magnetic	Friction, viscous

TRICK - “Conservative = Coordinates Only + Closed Zero”

Path independent; closed-path work zero.

EXAM TRAP -

Source notes list gravitation, electrostatic and magnetic force under conservative examples.

4. Power — Work per Time

- Power = rate of doing work

- $P = W/t$
- Scalar quantity
- SI unit = Watt (W) = J/s
- 1 HP = 746 W
- 1 kW = 10^3 W; 1 MW = 10^6 W

TRICK - “Power = Work / Time; 1 HP = 746 W”

746 ko 1000 se confuse mat karo.

TRICK - “kW = 10^3 ; MW = 10^6 ”

Kilo 3 zeros; Mega 6 zeros.

EXAM TRAP -

1 Horse Power = 746 W, not 1000 W.

5. Energy — Capacity to Work

- Energy = capacity to do work
- Scalar quantity
- SI unit = Joule
- Commercial unit of electrical energy = kilowatt-hour (kWh)

EXAM TRAP - “Work aur Energy dono Joule; Commercial Electricity = kWh”

kWh energy unit hai, power unit nahi.

6. Energy Unit Pack — erg / cal / kWh / eV

Unit	Joule Equivalent
erg	10^{-7} J
Calorie	4.2 J
kWh	3.6×10^6 J
eV	1.6×10^{-19} J

TRICK - “7-4.2-3.6-1.6”

erg 10^{-7} J; calorie 4.2 J; kWh 3.6×10^6 J; eV 1.6×10^{-19} J.

EXAM TRAP -

1 kWh = 3.6×10^6 J; 1 eV = 1.6×10^{-19} J.

7. Forms of Energy — T-E-C-L-S-N-M

Form	Memory
Thermal	Heat
Electric	Flow of electric current
Chemical	Chemical reactions
Light	Sun / light sources
Sound	Sound sources / instruments
Nuclear	Fission or fusion
Mechanical	KE + PE

TRICK - “T-E-C-L-S-N-M”

Thermal-Electric-Chemical-Light-Sound-Nuclear-Mechanical.

TRICK - “Mechanical = KE + PE”

Source chapter ka direct definition.

8. Kinetic Energy — Motion + v^2

- Kinetic energy = energy due to motion
- $KE = \frac{1}{2} mv^2$
- Momentum relation: $KE = \frac{p^2}{2m} = \frac{1}{2} pv$
- Examples: wind mills; fired bullet
- Kinetic energy always positive in source notes

TRICK - “KE = Half m v-Square”

Velocity double ho to v^2 relation yaad rahe.

EXAM TRAP - “Same Momentum → Lighter Mass = Higher KE”

$KE = \frac{p^2}{2m}$, so equal p par m kam → KE zyada.

9. Potential Energy — Height vs Spring

Type	Formula	Examples
Gravitational PE	$U = mgh$	Body at height
Elastic PE	$U = \frac{1}{2} kx^2$	Stretched bow; compressed spring

TRICK - “Height = mgh ; Spring = $\frac{1}{2} kx^2$ ”

Gravitational vs elastic PE formula pair.

TRICK - “k = spring constant; x = extension/deformation”

Elastic-energy symbols lock karo.

10. KE vs PE — Motion vs Position

Feature	Kinetic	Potential
Depends on	Mass + velocity	Mass + height/configuration
Formula	$\frac{1}{2} mv^2$	mgh or $\frac{1}{2} kx^2$
Example	Moving bullet; running car	Dam water; stretched spring
Can be zero?	Yes, $v = 0$	Yes, reference $h = 0$

TRICK - “KE = Move; PE = Place”

Motion vs position/configuration.

11. Conservation of Energy — Total Constant

- Energy neither created nor destroyed; only converted from one form to another
- Total energy of isolated system remains constant
- Ideal free fall: $KE + PE = \text{Total Mechanical Energy} = \text{Constant}$

TRICK - “Energy Changes Form, Not Total”

Conservation law ka one-line core.

12. Free Fall — Top/Mid/Bottom

Position	KE	PE	Total
Highest	Minimum (0)	Maximum	Constant
Midway	1/2 total	1/2 total	Constant
Lowest	Maximum	Minimum (0)	Constant

TRICK - “Top = PE; Bottom = KE; Mid = Half-Half”

Top PE max/KE0; bottom KE max/PE0.

EXAM TRAP -

Total mechanical energy stays constant in the source free-fall model.

13. Energy Conversion — Opposite Pairs

Device	Conversion
Solar cell	Solar/Light → Electrical
Dynamo	Mechanical → Electrical
Electric Motor	Electrical → Mechanical
Microphone	Sound → Electrical
Loudspeaker	Electrical → Sound
Sitar	Mechanical → Sound
Bulb / Tube light / Heater	Electrical → Light + Thermal
Burning candle	Chemical → Light + Thermal
Burning coal	Chemical → Thermal
Battery / Electric cell	Chemical → Electrical
Engine	Thermal → Mechanical
Photoelectric cell	Light → Electrical

TRICK - “Dynamo <-> Motor”

Dynamo: Mechanical → Electrical.

Motor: Electrical → Mechanical.

TRICK - “Microphone <-> Loudspeaker”

Microphone: Sound → Electrical.

Loudspeaker: Electrical → Sound.

EXAM TRAP - “Solar Cell + Photoelectric Cell = Light → Electrical”

Source notes dono ko same conversion se link karte hain.

14. Device Chain — C-B-E-S

TRICK - “Chemical Battery → Electrical Bulb → Light/Thermal; Thermal Engine → Mechanical”

Battery = Chemical→Electrical.

Bulb/heater = Electrical→Light+Thermal.

Engine = Thermal→Mechanical.

Sitar = Mechanical→Sound.

15. Source Worked Examples — Number Hooks

Example	Source Result
F=40N, s=10m, $\theta=30^\circ$	W = 346.41 J
m=40kg, h=2m, t=4s	Work 784 J; Power 196 W

$m=1500\text{kg}$, $v=72\text{km/h}=20\text{m/s}$	$\text{KE} = 3 \times 10^5 \text{ J}$
$m=40\text{kg}$, $h=5\text{m}$, $g=10\text{m/s}^2$	$\text{PE} = 2000 \text{ J}$

TRICK - “346.41 / 196 / 3×10^5 / 2000”

Source ke four worked-example answers ka recall pack.

16. Final Exam Traps — 10 Red Flags

- Work scalar; $W = F_s \cos\theta$; sign $\cos\theta$ decides
- Circular motion: force perpendicular displacement $\rightarrow$ work = 0
- $1 \text{ J} = 10^7 \text{ erg}$; $1 \text{ erg} = 10^{-7} \text{ J}$
- Power = W/t ; SI unit Watt = J/s
- $1 \text{ HP} = 746 \text{ W}$, not 1000 W
- $1 \text{ calorie} = 4.2 \text{ J}$; $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$; $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$
- $\text{KE} = 1/2 mv^2$; equal momentum par lighter body has higher KE
- Gravitational $\text{PE} = mgh$; elastic $\text{PE} = 1/2 kx^2$
- Highest point: PE max/KE0 ; lowest: KE max/PE0
- Dynamo $\text{M} \rightarrow \text{E}$; Motor $\text{E} \rightarrow \text{M}$; Microphone $\text{S} \rightarrow \text{E}$; Loudspeaker $\text{E} \rightarrow \text{S}$

EXAM TRAP - “ $\cos\theta$ - Circle - J/erg - Watt - 746 - Units - KE - PE - FreeFall - Devices”

Chapter ke 10 red flags isi order mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Work $W = F_s \cos\theta$ scalar; J SI, erg CGS, $1\text{J}=10^7\text{erg}$ $\rightarrow$ sign: same +, 90° zero, opposite -; falling gravity +, lifting gravity -, circular work 0 $\rightarrow$ Conservative path-independent/closed 0 vs Non-conservative path-dependent $\rightarrow$ Power $P = W/t$ scalar, Watt= J/s , HP746W, $\text{kW}10^3$, $\text{MW}10^6 \rightarrow$ Energy capacity-to-work; commercial kWh; units $\text{erg}10^{-7}\text{J}$, cal4.2J, $\text{kWh}3.6 \times 10^6\text{J}$, $\text{eV}1.6 \times 10^{-19}\text{J} \rightarrow$ forms T-E-C-L-S-N-M; Mechanical= $\text{KE}+\text{PE} \rightarrow \text{KE}=1/2 mv^2=p^2/2m=1/2 pv$; same p lighter mass higher KE $\rightarrow$ PE: height mgh, spring $1/2 kx^2 \rightarrow$ conservation: total isolated constant; free fall Top PE, Mid half-half, Bottom KE $\rightarrow$ Conversions: Solar/Photoelectric Light $\rightarrow$ Electrical; Dynamo $\text{M} \rightarrow \text{E} \leftrightarrow$ Motor $\text{E} \rightarrow \text{M}$; Microphone $\text{S} \rightarrow \text{E} \leftrightarrow$ Loudspeaker $\text{E} \rightarrow \text{S}$; Sitar $\text{M} \rightarrow \text{S}$; Battery $\text{C} \rightarrow \text{E}$; Engine $\text{T} \rightarrow \text{M}$; Candle $\text{C} \rightarrow \text{L}+\text{T}$.”

TOP 5: Circular work=0 | HP=746 W | kWh= $3.6 \times 10^6 \text{ J}$ | Equal momentum: lighter $\rightarrow$ higher KE | Dynamo $\leftrightarrow$ Motor & Microphone $\leftrightarrow$ Loudspeaker.

GRAVITATION — MEMORY TRICK SHEET

General Science · Physics · Chapter 5 · Newton Law + G/g + Kepler + Satellites + Escape Velocity + Mass/Weight

Ye full notes nahi hai — source chapter ke G vs g, g-variation, Kepler laws, orbital/escape velocity, geostationary-polar satellites aur mass-weight traps ko fast exam-recall format mein compress kiya gaya hai.

1. Gravitation — Newton + Cavendish

- Gravitation = universe ke kisi bhi do bodies ke beech attraction force
- Universal law formulated by Isaac Newton
- Universal gravitational constant G experimentally determined by Henry Cavendish in 1798

TRICK - “Newton Law; Cavendish G-1798”

Newton = law.

Cavendish = G ka experimental determination.

EXAM TRAP -

Law aur G-determination ke scientists ko interchange mat karna.

2. Universal Law — Product Up, Distance-Square Down

- $F \propto m_1 m_2 / r^2$
- $F = G(m_1 m_2) / r^2$
- Force masses ke product ke directly proportional
- Centres ke distance ke square ke inversely proportional

TRICK - “Mass Product UP; r^2 DOWN”

$m_1 m_2$ numerator; r^2 denominator.

3. G vs g — Fixed vs Variable

Feature	G	g
Meaning	Universal gravitational constant	Gravitational acceleration
Value	$6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$	Earth near surface $\approx 9.8 \text{ m/s}^2$
Nature	Universal / fixed	Changes with location
Depends on location?	No	Yes

TRICK - “Capital G = Global Fixed; small g = Geography Changes”

G everywhere same.

g location ke saath change hota hai.

EXAM TRAP - “G = 6.67×10^{-11} ; g = 9.8”

Value pair.

4. g Formula + Core Values

- $g = G M_e / R_e^2$
- Earth surface $\approx 9.8 \text{ m/s}^2$
- Moon surface $\approx 1/6$ of Earth $\approx 1.63 \text{ m/s}^2$
- Earth centre = 0
- g falling body ke mass par depend nahi karta

TRICK - “Earth 9.8; Moon 1/6; Centre 0”

Three-value chain.

EXAM TRAP -

g does not depend on mass of falling body.

5. g Variation — P-E-Up-Down-Centre

Location	g	Why / Memory
Poles	Maximum	Closer to centre + rotation effect minimum
Equator	Minimum	Farther + rotational effect maximum
Altitude increases	Decreases	Distance from centre increases
Depth increases	Decreases	Effective attracting mass reduces
Earth centre	0	Net pull cancels

TRICK - “Pole Peak; Equator Low; Up Down Both g Down; Centre Zero”

$P > E$; altitude $\uparrow g \downarrow$; depth $\uparrow g \downarrow$; centre 0.

TRICK - “Altitude $\approx 1/r^2$; Depth $\approx$ Linear Down”

Source recall: altitude with distance inverse-square; depth approx linearly to zero.

EXAM TRAP -

g maximum poles, minimum equator — reverse mat karna.

6. Kepler’s 3 Laws — E-A-T

Law	Name	Key Point
1	Law of Orbits	Elliptical orbit; Sun at one focus
2	Law of Equal Areas	Equal areas in equal times; faster perihelion, slower aphelion
3	Harmonic Law	$T^2 \propto a^3$

TRICK - “1-Ellipse, 2-Areas, 3- T^2a^3 ”

Kepler = E-A-T.

TRICK - “Peri = Fast; Aphe = Slow”

Second-law speed trap.

EXAM TRAP - “Sun at FOCUS, not Centre”

Law 1 ka common MCQ trap.

TRICK - “Law 2 = Constant Areal Speed = Angular Momentum Link”

Source notes Law 2 ko conservation of angular momentum se connect karte hain.

7. Satellite Basics — Free Fall + Weightlessness

- Satellite = body revolving around a planet
- Moon = natural satellite; INSAT/GPS-type spacecraft = artificial satellites
- Orbital velocity near Earth $V_o = \sqrt{gR} \approx 8 \text{ km/s}$
- Satellite in orbit continuously free fall mein hota hai
- Apparent weight in orbit = 0; mass unchanged

TRICK - “Orbit = Free Fall; Weight 0, Mass Same”

Weightlessness means apparent weight zero, mass zero nahi.

EXAM TRAP -

Mass unchanged during orbital weightlessness.

8. Geostationary — 24 / 36,000 / Equator / W→E

Feature	Geostationary
Time period	24 h
Height	$\approx 36,000$ km above Earth
Orbital plane	Equatorial
Direction	West → East
Appearance	Stationary relative to Earth surface
Uses	TV, communication, weather monitoring
Indian example	INSAT series

TRICK - “Geo = 24-36-E-WE”

24 h, 36,000 km, Equator, West-to-East.

EXAM TRAP - “36,000 not 3,600”

Zero count ka direct exam trap.

9. Geostationary vs Polar

Feature	Geostationary	Polar
Orbit	Equatorial	Near poles
Height	$\approx 36,000$ km	≈ 500 -800 km
Period	24 h	Much shorter
Main use	Communication/weather/TV	Remote sensing/mapping/surveillance

TRICK - “GEO = Communication; POLAR = Picture/Mapping”

Use-based distinction.

10. Orbital vs Escape Velocity — 8 vs 11.2

Quantity	Formula	Earth Value
Orbital velocity V_o	$\sqrt{gR}$	≈ 8 km/s
Escape velocity V_e	$\sqrt{2gR}$	11.2 km/s
Relation	$V_e = \sqrt{2} V_o$	$\approx 41\%$ higher

TRICK - “Orbit 8; Escape 11.2; $V_e = \sqrt{2} V_o$ ”

8 → 11.2 relation pack.

TRICK - “Escape Earth 11.2; Moon 2.35”

Earth-Moon escape value pair.

EXAM TRAP -

Escape velocity projectile mass par depend nahi karta; planet mass/radius par karta hai.

11. Escape Speed Meaning

- Minimum launch speed needed to escape planet gravitational field permanently without further propulsion
- Below 11.2 km/s from Earth: body eventually returns
- At 11.2 km/s: just escapes
- Above it: escapes with excess kinetic energy

TRICK - “Below Return; Equal Just Escape; Above Extra KE”

Earth escape threshold logic.

12. Mass vs Weight — m vs mg

Feature	Mass	Weight
Meaning	Amount of matter / intrinsic property	Gravitational force
Formula	—	$W = mg$
SI unit	kg	newton N
Type	Scalar	Vector
Changes with location?	No	Yes
Earth centre	Same	0
Moon	Same	1/6 Earth weight

TRICK - “Mass = Matter = Same; Weight = mg = g Changes”

Mass fixed; weight location dependent.

TRICK - “Moon: Mass Same, Weight 1/6”

Classic exam pair.

EXAM TRAP -

Weightlessness = apparent weight 0, mass 0 nahi.

13. Balance Trap — Spring vs Beam

- Spring balance measures weight
- Beam balance compares mass

EXAM TRAP - “Spring = Weight; Beam = Mass”

Instrument pair.

14. Formula + Value Flash Table

Item	Recall
G	$6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$
g Earth	9.8 m/s^2
g Moon	$\approx 1.63 \text{ m/s}^2 \approx g/6$
g Earth centre	0
Vo Earth	$\approx 8 \text{ km/s}$
Ve Earth	11.2 km/s
Ve Moon	2.35 km/s
Geo height / period	$\approx 36,000 \text{ km} / 24 \text{ h}$
Newton gravitation	$F = Gm_1m_2/r^2$
Weight	$W = mg$
Kepler 3	$T^2 \propto a^3$
Orbital / Escape	$V_o = \sqrt{gR}$; $V_e = \sqrt{2gR} = \sqrt{2}V_o$

15. Final Exam Traps — 10 Red Flags

- Newton gave law; Cavendish determined G in 1798
- G fixed universal; g varies with location
- g poles maximum, equator minimum, centre zero
- Mass same everywhere; weight changes with g
- Kepler 1 ellipse; Sun at one focus, not centre
- Kepler 2 equal areas/equal times; perihelion faster, aphelion slower
- Satellite orbit = continuous free fall; apparent weight zero, mass same
- Geostationary = 24 h, $\approx 36,000$ km, equatorial, West $\rightarrow$ East
- Orbital ≈ 8 km/s; Escape Earth 11.2 km/s; $V_e = \sqrt{2}V_o$
- Escape velocity independent of projectile mass

EXAM TRAP - “Newton/Cavendish - G/g - $P/E/C$ - Mass/Weight - $K1/K2$ - Satellite - Geo - 8/11.2 - Mass-independent”

Chapter ke 10 highest-yield traps isi chain mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Newton gravitation law $F = Gm_1m_2/r^2$; Cavendish G -1798; $G = 6.67 \times 10^{-11}$ fixed vs $g = 9.8$ variable $\rightarrow g = GMe/R_e^2$; Moon $g \approx 1/6$, centre 0; Pole Peak, Equator Low, altitude $\uparrow g \downarrow$, depth $\uparrow g \downarrow \rightarrow$ Kepler E-A-T: 1 ellipse Sun focus, 2 equal areas/peri fast-aphe slow, 3 $T^2 \propto a^3$; law 2 angular-momentum link $\rightarrow$ satellite = continuous free fall; $V_o = \sqrt{gR} \approx 8$ km/s; weightlessness apparent weight 0, mass same $\rightarrow$ Geostationary 24h + 36,000km + equator + West $\rightarrow$ East + TV/communication/weather; Polar 500-800km + remote sensing/mapping $\rightarrow$ Escape $V_e = \sqrt{2gR} = \sqrt{2}V_o$; Earth 11.2, Moon 2.35; below-return/equal-just-escape/above-extra-KE; independent of projectile mass $\rightarrow$ Mass matter scalar kg same; Weight $W = mg$ vector N varies; centre weight 0, Moon weight 1/6; Spring balance weight, Beam balance mass.”

TOP 5: G fixed vs g variable | Pole max-Equator min-Centre 0 | Kepler 1/2/3 = Ellipse/Areas/ $T^2 \propto a^3$ | Geo 24h-36,000km | Orbit 8 vs Escape 11.2 and $V_e = \sqrt{2}V_o$.

MECHANICAL PROPERTIES OF MATTER — MEMORY TRICK SHEET

General Science · Physics · Chapter 6 · Elasticity + Pressure + Fluids + Flotation + Surface Tension + Viscosity + Bernoulli

Ye full notes nahi hai — source chapter ke stress-strain, elastic moduli, atmospheric/fluid pressure, Pascal, Archimedes, surface tension, capillarity, viscosity aur Bernoulli ke high-memory facts ko exam-recall tricks mein compress kiya gaya hai.

1. Matter States — S-L-G-P-B

State	Mechanical Hook
Solid	Elasticity; definite shape + volume
Liquid	Pressure, flotation, surface tension, capillarity, viscosity
Gas	Atmospheric pressure
Plasma	4th state
Bose-Einstein Condensate	5th state

TRICK - “S-L-G-P-B = Solid-Liquid-Gas-Plasma-BEC”

Plasma = 4th; Bose-Einstein Condensate = 5th.

EXAM TRAP -

Plasma 4th, BEC 5th — order ulta mat karna.

2. Stress vs Strain — Unit vs No Unit

- Stress = restoring force per unit area; Stress = F/A
- Stress SI unit = $N/m^2 = Pa$
- Strain = change in dimension / original dimension
- Strain dimensionless hai; no unit

Stress Type	Meaning
Longitudinal-Compressive	Length decreases
Longitudinal-Tensile	Length increases
Normal Stress	Force perpendicular to surface
Shear Stress	Force tangential to surface

Strain Type	Formula	Meaning
Longitudinal / Linear	$\Delta L/L$	Length change
Volumetric	$\Delta V/V$	Volume change
Shearing	$\Delta x/L$	Angular deformation

EXAM TRAP - “Stress has Pa; Strain has NO unit”

Dono ka sabse common exam confusion.

3. Hooke + Elastic Moduli — Y- β - η

- Hooke’s Law: elastic limit ke andar Stress $\propto$ Strain
- Elastic modulus E = Stress/Strain
- Modulus unit = N/m^2 ; dimension = $[ML^{-1}T^{-2}]$
- Law named after Robert Hooke

Modulus	Ratio / Formula	Best Memory
Young’s Y	Longitudinal Stress / Longitudinal Strain; $Y = MgL/(\pi r^2 \Delta L)$	Length
Bulk β	Normal Stress / Volumetric Strain; $\beta = -PV/\Delta V$	Volume
Rigidity η	Shear Stress / Shear Strain; $\eta = F/(A\theta) = FL/(A\Delta x)$	Shape

TRICK - “Y = Length; β = Volume; η = Shape”

Y- β - η ko Length-Volume-Shape ke saath lock karo.

TRICK - “Liquids/Gases = Bulk Only”

Source ke mutabik liquids aur gases ke paas Young's ya rigidity modulus nahi.

EXAM TRAP -

Bulk modulus mein negative sign compression ke under volume decrease ki wajah se.

4. Thrust vs Pressure

Feature	Thrust	Pressure
Meaning	Force perpendicular to surface	Thrust per unit area
Formula	—	$P = F/A$
SI Unit	N	$\text{Pa} = \text{N/m}^2$

TRICK - “Thrust = Force; Pressure = Force/Area”

Pressure area ke saath inverse effect dikhata hai for same force.

5. Atmospheric Pressure — Barometer + Altitude

- $1 \text{ atm} \approx 10^5 \text{ Pa}$
- Atmospheric pressure measured by barometer
- Altitude badhne par atmospheric pressure decreases

Barometer Trend	Weather
Falls suddenly	Storm
Falls slowly	Rain
Rises slowly	Clear weather

TRICK - “Sudden Fall = Storm; Slow Fall = Rain; Slow Rise = Clear”

Barometer weather chain.

TRICK - “High Altitude: Pressure ↓ + Boiling Point ↓ + Cooking Time ↑”

Pressure cooker opposite: pressure ↑ → boiling point ↑ → cooking faster.

EXAM TRAP -

High altitude par pressure aur boiling point dono decrease.

6. Density + Relative Density

Quantity	Formula / Value	Exam Hook
Density ρ	M/V	SI kg/m^3
Relative Density	$\rho_{\text{substance}} / \rho_{\text{water at } 4^\circ\text{C}}$	No unit; hydrometer
Pure Water	10^3 kg/m^3	Maximum density at 4°C
Ice	0.9 g/cm^3	Less dense than water

TRICK - “Water = 1000 + Max at 4°C ; Ice = 0.9”

Density number chain.

EXAM TRAP -

Relative density has no unit; source notes hydrometer as instrument.

7. Hydrostatic Pressure + Pascal

- Pressure at depth h : $P = h\rho g$
- Total pressure: $P_{\text{total}} = P_a + h\rho g$

- Source example at 10 m water depth: total $\approx 2.01 \times 10^5 \text{ Pa} \approx 2 \text{ atm}$

TRICK - “Depth $\uparrow$ $\rightarrow$ Pressure $\uparrow$: $h\rho g$ ”

Hydrostatic pressure formula.

- Pascal’s Law: enclosed fluid par applied pressure equally every direction transmit hota hai
- Applications: Hydraulic Lift, Hydraulic Brakes, Hydraulic Press

TRICK - “Pascal = Lift-Brake-Press”

Hydraulic trio.

EXAM TRAP -

Hydraulic lift/brakes/press = Pascal, Bernoulli nahi.

8. Archimedes + Flotation

- Immersed body experiences upward buoyant force equal to weight of displaced fluid
- Buoyant force / upthrust = weight of displaced fluid
- Apparent weight in fluid = actual weight - buoyant force
- Applications: life preservers, submarines, lactometers

Condition	Result	Examples
Weight > Buoyant force	Sinks	Iron nail
Weight = Buoyant force	Equilibrium / submerged float	Submarine, neutral body
Weight < Buoyant force	Rises / partly floats	Ship, wood block

TRICK - “W>B Sink; W=B Balance; W<B Rise”

Flotation condition chain.

TRICK - “Archimedes = Upthrust = Displaced Fluid Weight”

Principle ka one-line core.

9. Floating Ice — 1/10 Up, 9/10 Down

- Floating ice ka about 1/10 water ke above aur 9/10 submerged hota hai
- Floating ice completely melt hone par water level unchanged rehta hai
- Lactometer aur hydrometer flotation principle par work karte hain

EXAM TRAP - “Ice = 1 Up : 9 Down; Melt = Level Same”

Iceberg trap ko number ke saath yaad karo.

10. Surface Tension — F/l

- Surface tension = force per unit length in liquid surface plane
- $T = F/l = \text{mg/l}$
- Unit = N/m
- Numerically equivalent surface-energy unit = J/m^2
- Surface energy = $W/\Delta A$

TRICK - “Surface Tension = Force/Length; Surface Energy = Work/Area”

N/m vs J/m^2 pair.

Condition	Effect
Temperature increases	Surface tension decreases
Critical temperature	Surface tension = 0
Soap added	Surface tension decreases
Kerosene spread	Surface tension decreases

TRICK - “Heat ↑ → ST ↓; Critical → 0”

Temperature trend.

- Wet brush hairs stick together → surface tension
- Needle can float on water → surface tension
- Raindrops spherical → surface tension minimizes surface area

11. Capillarity — Thin Tube, More Rise

- Capillarity = narrow tube mein liquid ka rise/fall due to surface tension
- Examples: plant roots/stems water transport, lamp wick oil rise, wetting, fine capillary movement
- Capillary rise $\propto 1 / \text{tube radius}$

EXAM TRAP - “Radius Thin → Rise High”

Capillary rise inverse to tube radius.

12. Viscosity — Fluid Internal Friction

- Viscosity adjacent fluid layers ke relative motion ko oppose karne wali internal friction hai
- Property of both liquids and gases
- $F = -\eta A(\Delta v / \Delta z)$
- $\eta = F \Delta z / (A \Delta v)$
- SI unit = $\text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-1}$; also called poiseuille (PI)

TRICK - “Viscosity = Fluid Friction”

Solids ki property nahi; liquids + gases.

High → Low Viscosity	Order
Typical order	Honey > Water > Kerosene > Petrol > Gas

TRICK - “H-W-K-P-G”

Honey > Water > Kerosene > Petrol > Gas.

13. Temperature vs Viscosity — Opposite Trend

Substance	Temperature ↑ → Viscosity
Liquids	Decreases
Gases	Increases

EXAM TRAP - “Liquid Low; Gas Grow”

Temperature badhne par Liquid viscosity ↓, Gas viscosity ↑.

14. Fluid Flow — Laminar vs Turbulent

Flow	Nature
Streamline / Laminar	Orderly regular paths; usually low velocity
Turbulent	Irregular chaotic paths; above critical velocity
Critical Velocity	Maximum velocity up to which streamline flow persists

TRICK - “Laminar = Lines; Turbulent = Trouble”

Ordered vs chaotic.

15. Bernoulli — Fast Flow = Low Pressure

- Ideal incompressible fluid in streamline flow ke liye total energy per unit volume constant
- $P + \rho gh + \frac{1}{2} \rho v^2 = \text{constant}$
- Based on conservation of energy

- Fast flow → low pressure; slow flow → high pressure

TRICK - “Bernoulli = Fast LOW, Slow HIGH”

Velocity-pressure inverse recall.

Application	Source Logic
Aeroplane Wings	Faster air above → lower pressure → lift
Venturimeter	Narrow section pressure difference → flow speed
Magnus Effect	Spinning ball pressure difference → curve
Tin Roof	Fast wind above → low pressure → upward lift
Train Platform Suction	Fast train lowers nearby pressure

TRICK - “A-V-M-T-T = Bernoulli Applications”

Aeroplane - Venturimeter - Magnus - Tin roof - Train platform.

EXAM TRAP -

Bernoulli applies to ideal incompressible fluid in streamline flow.

16. Formula Flash

Quantity / Law	Formula / Unit
Stress	F/A ; N/m^2
Strain	$\Delta L/L$, $\Delta V/V$, $\Delta x/L$; no unit
Elastic Modulus	Stress/Strain ; N/m^2
Pressure	F/A ; Pa
Hydrostatic Pressure	$h\rho g$
Density	M/V ; kg/m^3
Surface Tension	F/l ; N/m
Surface Energy	$W/\Delta A$; J/m^2
Viscous Force	$-\eta A(\Delta v/\Delta z)$
Bernoulli	$P + \rho gh + 1/2\rho v^2 = \text{constant}$

17. Final Exam Traps — 10 Red Flags

- Strain dimensionless hai; no unit
- Plasma 4th state; Bose-Einstein Condensate 5th
- Liquids/gases have bulk modulus, not Young’s or rigidity modulus
- High altitude: pressure ↓ and boiling point ↓, so cooking longer
- Water maximum density at 4°C; ice less dense
- Hydraulic lift/brakes/press = Pascal’s Law
- Buoyant force = weight of displaced fluid
- Floating ice melts → water level unchanged
- Liquid viscosity ↓ with temperature; gas viscosity ↑
- Aeroplane lift + Venturimeter = Bernoulli applications

EXAM TRAP - “Strain-P/BEC-Bulk-Altitude-Water-Pascal-Buoyancy-Ice-Viscosity-Bernoulli”

Chapter ke 10 core traps isi order mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Matter S-L-G-P-B: Plasma4, BEC5 → Stress F/A Pa vs Strain ratio no-unit; stress types compress/tensile/normal/shear; strain linear/volume/shear → Hooke stress $\propto$ strain; moduli Y -length, β -volume, η -shape; liquids+gases bulk only; β negative in compression → Thrust force, Pressure F/A ; 1 atm $\approx 10^5$ Pa; barometer sudden fall storm, slow fall rain, slow rise clear; altitude pressure $\downarrow$ + boiling point $\downarrow$ → Density M/V ; water 1000 max at 4°C ; ice 0.9; Hydro $P=h\rho g$; Pascal = lift-brake-press → Archimedes upthrust=displaced-fluid weight; $W>B$ sink, $W=B$ balance, $W<B$ rise; ice 1/10 up-9/10 down, melt level same → Surface Tension F/l N/m, Surface Energy $W/\Delta A$ J/m²; heat $\uparrow$ ST $\downarrow$; capillary thin tube $\rightarrow$ higher rise → Viscosity fluid friction, $H>W>K>P>G$; liquid temp $\uparrow$ viscosity $\downarrow$, gas $\uparrow$ → laminar orderly vs turbulent chaotic → Bernoulli $P+\rho gh+1/2\rho v^2$ constant, conservation of energy; Fast flow LOW pressure; Aeroplane-Venturi-Magnus-Tin roof-Train platform.”

TOP 5: Stress Pa vs Strain no-unit | Y -Length β -Volume η -Shape | Pascal hydraulic trio | Ice melt level same | Liquid viscosity $\downarrow$ but Gas viscosity $\uparrow$ with heat.

TEMPERATURE, HEAT & THERMODYNAMICS — MEMORY TRICK SHEET

General Science · Physics · Chapter 7 · Scales + Thermometers + Heat + Expansion + State Change + Heat Transfer + Thermodynamic Laws

Ye full notes nahi hai — source chapter ke temperature-scale conversions, thermometer ranges, specific/latent heat, thermal expansion, water anomaly, phase-change effects, heat-transfer methods aur thermodynamic laws ko fast exam-recall format mein compress kiya gaya hai.

1. Temperature vs Heat — Measure vs Transfer

Concept	Core Idea	Unit / Hook
Temperature	Hotness/coldness ka measure	Kelvin K; thermometer
Heat	Temperature difference ke kaaran transferred energy	Joule J; hot > cold
Thermodynamics	Heat-mechanical energy relation	P, V, T variables

TRICK - “Temperature = Measure; Heat = Transfer”

Heat naturally hotter body se colder body ki taraf flow karta hai.

EXAM TRAP -

Temperature aur heat same cheez nahi: temperature measure hai, heat transferred energy.

2. Temperature Scales — C-F-R-K-Ra

Scale	Freezing	Boiling	Divisions
Celsius	0°C	100°C	100
Fahrenheit	32°F	212°F	180
Reaumur	0°R	80°R	80
Kelvin	273 K	373 K	100
Rankin	492 Ra	672 Ra	180

TRICK - “C-F-R-K-Ra = 100-180-80-100-180”

Scale order ke saath interval divisions lock karo.

TRICK - “Universal Relation”

$(C-0)/100 = (F-32)/180 = (R-0)/80 = (K-273)/100 = (Ra-492)/180$.

TRICK - “60°F = 288.5 K”

Source solved conversion example ka final answer.

3. Fixed Temperature Facts — 0 / -40 / 273.16

Fact	Source Value
Absolute zero	-273.15°C = 0 K
Celsius = Fahrenheit	-40°C = -40°F
Fahrenheit = Kelvin	574.25°F = 574.25 K
Fahrenheit = Reaumur	-25.6°F = -25.6°R
Human body	37°C = 98.4°F = 310.5 K
Triple point of water	273.16 K

TRICK - “-40 = C and F Same”

Celsius-Fahrenheit ka only equal reading source notes mein -40.

EXAM TRAP - “Absolute 0K; Triple 273.16K”

Absolute zero aur triple-point pair.

4. Thermometers — Range Memory

Thermometer	Range	Special Use
Gas (Nitrogen)	500°C to 1500°C	Very high temperatures
Platinum	-200°C to 1000°C	Wide range, precision
Thermocouple	-200°C to 1600°C	Widest listed range
Total Radiation	Above 800°C	Very high temp, no contact
Alcohol	Below -40°C; fp $\approx$ -115°C	Very low temperatures
Clinical	35°C to 42°C	Human body

TRICK - “Clinical 35-42; Thermocouple -200 to 1600”

Source ke two highest-frequency thermometer ranges.

TRICK - “Alcohol for Very Low; Pyrometer No Contact”

Alcohol preferred because mercury freezes near -39°C.

Total-radiation thermometer very high temperature without contact.

EXAM TRAP -

Widest listed thermometer range = Thermocouple, source table ke mutabik.

5. Heat Units — J-cal-BTU

Item	Value / Unit
SI heat unit	Joule J
CGS heat unit	Calorie cal
British unit	BTU
1 BTU	252 cal
Mechanical equivalent	1 cal = 4.186 J

TRICK - “1 cal = 4.186 J; 1 BTU = 252 cal”

4.186 aur 252 ko number pair mein yaad karo.

EXAM TRAP -

BTU = 252 cal, not 1000 cal.

6. Specific Heat — $Q = ms\Delta T$

- Specific heat = unit mass ka temperature 1°C raise karne ke liye required heat
- $s = Q/(m\Delta T)$, hence $Q = ms\Delta T$
- SI unit = J/kg°C; CGS = cal/g°C
- Water specific heat = 4186 J/kg°C = 1 cal/g°C
- Source uses: hot-water bottle, crop frost protection, vehicle radiator cooling

TRICK - “Specific = $Q / m\Delta T$; Water = 4186 = 1 cal/g°C”

Water ka high specific heat applications se jodo.

TRICK - “2 kg, 30→100, s=300 => $Q=4.2\times 10^4$ J”

Source solved example answer.

7. Latent Heat — State Changes, Temperature Same

- Latent heat = state change during absorbed/released heat without temperature change
- $L = Q/m$
- SI = J/kg; CGS = cal/g

Type	Change	Value
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Fusion	Solid > Liquid	80 cal/g
Vaporisation	Liquid > Gas	539 cal/g

TRICK - “Fusion 80; Vapor 539”

Vaporisation latent heat much larger than fusion.

EXAM TRAP - “Phase Change = Heat Changes, Temperature Doesn’t”

State-change ka most important conceptual trap.

TRICK - “Steam Burn More Severe”

Source reason: vaporisation latent heat 539 cal/g > fusion 80 cal/g.

8. Calorimetry — Lost = Gained

- Heat Lost by Hot Object = Heat Gained by Cold Object
- Based on conservation of thermal energy in an isolated system
- Used to determine unknown specific heats
- Ideal assumption: surroundings ko heat loss nahi

TRICK - “Hot Lost = Cold Gained”

Calorimetry ka one-line equation.

9. Thermal Expansion — $\alpha : \beta : \gamma = 1 : 2 : 3$

Expansion	Symbol	Formula	Dimension Change
Linear	α	$\Delta L / (L \Delta t)$	Length
Area/Superficial	β	$\Delta A / (A \Delta t)$	Area
Volumetric	γ	$\Delta V / (V \Delta t)$	Volume

TRICK - “ α - β - $\gamma = 1$ - 2 - 3 ”

$\beta = 2\alpha$; $\gamma = 3\alpha$.

TRICK - “Railway Gaps = Summer Expansion”

Thermal-expansion application.

EXAM TRAP -

$\alpha : \beta : \gamma = 1 : 2 : 3$ is a direct high-frequency source trap.

10. Anomalous Water — 0 to 4 Contracts

Temperature	Behaviour
$0^\circ\text{C} > 4^\circ\text{C}$	Water contracts; volume decreases
At 4°C	Volume minimum; density maximum
Above 4°C	Water expands; volume increases

TRICK - “0→4 Contract; 4 = Density Max; >4 Expand”

Water ka anomalous sequence.

TRICK - “Pond: 4°C Water Sinks, Surface Can Freeze”

Densest 4°C water sinks; lower water liquid reh sakta hai.

EXAM TRAP -

Water densest at 4°C , not 0°C .

11. Expansion Order — Gas > Liquid > Solid

- Gases show greatest volume expansion: Gases > Liquids > Solids
- Ideal gas volumetric expansion coefficient in source: $\gamma_p = \gamma_v = 1/273.15$ per °C
- Idealized relation mein all ideal gases same coefficient show karte hain

TRICK - “G-L-S = Greatest to Smallest Expansion”

Gas > Liquid > Solid.

12. State Changes — Absorb vs Release

Process	Transition	Heat
Melting	Solid > Liquid	Absorbed
Freezing	Liquid > Solid	Released
Boiling/Evaporation	Liquid > Gas	Absorbed
Condensation	Gas > Liquid	Released
Sublimation	Solid > Gas	Absorbed
Deposition	Gas > Solid	Released

TRICK - “Up in State = Absorb; Down in State = Release”

Solid→Liquid→Gas direction absorbs; reverse releases.

TRICK - “Sublime: Dry Ice-Camphor-Naphthalene-Iodine”

D-C-N-I source examples.

13. Pressure + Impurity Effects

Action	Melting Point	Boiling Point
Increase pressure	Decreases for ice	Increases
Decrease pressure	Increases	Decreases
Add impurity	Decreases	Increases

TRICK - “Pressure Cooker: $P \uparrow \rightarrow BP \uparrow \rightarrow$ Faster Cooking”

High altitude opposite: $P \downarrow \rightarrow BP \downarrow \rightarrow$ slower cooking.

EXAM TRAP - “Impurity: MP Down, BP Up”

Salt lowers freezing/melting point in source notes.

TRICK - “Triple Point = 273.16 K”

Solid, liquid and vapour coexist in equilibrium.

14. Heat Transfer — C-C-R

Method	How	Medium / State	Examples
Conduction	Direct contact transfer	Mainly solids; also mercury	Hot pan handle, soldering iron
Convection	Bulk particle motion	Liquids + gases	Sea/land breeze, hot-water circulation
Radiation	Electromagnetic waves	No medium; vacuum works	Sunlight, infrared heater

TRICK - “Conduction Solid; Convection Fluids; Radiation Vacuum”

C-C-R master rule.

EXAM TRAP - “Sea/Land Breeze = Convection”

Radiation nahi.

TRICK - “Thermos Cuts All 3”

Silvered walls reduce radiation; vacuum reduces conduction + convection.

15. Thermodynamics — $PV = \mu RT$

- Thermodynamics = heat aur mechanical energy ka relation
- System variables: pressure, volume, temperature
- Ideal gas equation in source notes: $PV = \mu RT$
- μ = number of moles, R = gas constant, T = absolute temperature

TRICK - “P-V-T System; $PV = \mu RT$ ”

Source notation μ for number of moles.

16. Zeroth, First, Second — Z-F-S

Law	Core Statement	Exam Key
Zeroth	A and B each thermal equilibrium with C => A and B equilibrium	Thermometry basis; Fowler (1931)
First	Heat supplied = ΔU + work done	$\Delta Q = \Delta U + \Delta W$; energy conservation
Second	100% heat→work impossible; cold→hot spontaneous flow impossible	No 100% efficient engine; refrigerator needs work

TRICK - “Z = Zero Temp Difference; F = First Energy; S = Second Direction”

Zeroth = thermal equilibrium.

First = conservation.

Second = direction/limits.

TRICK - “Fowler 1931 = Zeroth”

Source-specific scientist/year recall.

EXAM TRAP - “Cold → Hot Needs Work”

Refrigerator/AC external work consume karta hai.

17. Formula + Number Flash

Item	Recall
Scale conversion	$C/100 = (F-32)/180 = (K-273)/100$
Specific heat	$Q = ms\Delta T$
Latent heat	$L = Q/m$
Expansion	$\alpha:\beta:\gamma = 1:2:3$
First law	$\Delta Q = \Delta U + \Delta W$
Ideal gas	$PV = \mu RT$
1 calorie	4.186 J
1 BTU	252 cal
Water specific heat	4186 J/kg°C
Fusion / Vaporisation	80 / 539 cal/g

18. Final Exam Traps — 10 Red Flags

- Phase change ke during temperature constant rehta hai
- Radiation vacuum mein work karta hai; conduction/convection matter require karte hain
- Sea breeze + land breeze = convection
- Water maximum density at 4°C, not 0°C
- Ice floats because its density liquid water se lower hai
- Pressure cooker: pressure $\uparrow$ $\rightarrow$ boiling point $\uparrow$ $\rightarrow$ faster cooking
- High altitude: pressure $\downarrow$ $\rightarrow$ boiling point $\downarrow$ $\rightarrow$ slower cooking
- 1 BTU = 252 cal
- Clinical thermometer = 35°C-42°C; Thermocouple = widest listed -200°C to 1600°C
- Zeroth = Fowler/thermal equilibrium; First = conservation; Second = no 100% heat-to-work

EXAM TRAP - “Phase-Radiation-Breeze-Water-Ice-Pressure-Altitude-BTU-Thermometer-Laws”

Chapter ke 10 major traps isi sequence mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Temperature measure vs Heat transfer hot $\rightarrow$ cold $\rightarrow$ scales C-F-R-K-Ra with divisions 100-180-80-100-180; -40 C=F; absolute 0=0K; triple 273.16K; body 37°C=98.4°F=310.5K $\rightarrow$ Clinical 35-42; Thermocouple -200 to 1600 widest; Alcohol very low; total-radiation no contact $\rightarrow$ Heat units J/cal/BTU: 1 cal=4.186J, 1 BTU=252 cal $\rightarrow$ Specific $Q=ms\Delta T$, water 4186; Latent $L=Q/m$, Fusion 80 vs Vapor 539; phase change temp same; Calorimetry Hot Lost=Cold Gained $\rightarrow$ Expansion $\alpha:\beta:\gamma=1:2:3$; railway gaps; Water 0 $\rightarrow$ 4 contracts, 4°C density max, above 4 expands; expansion $G>L>S \rightarrow$ State up absorbs, down releases; sublimation D-C-N-I; Pressure $\uparrow$ BP $\uparrow$, altitude P $\downarrow$ BP $\downarrow$, impurity MP $\downarrow$ BP $\uparrow$; Triple 273.16K $\rightarrow$ Heat transfer C-C-R: conduction solids, convection fluids, radiation vacuum; sea/land breeze convection; thermos reduces all 3 $\rightarrow$ Thermo PV= μ RT; Zeroth equilibrium/Fowler 1931; First $\Delta Q=\Delta U+\Delta W$ conservation; Second 100% heat $\rightarrow$ work impossible and cold $\rightarrow$ hot needs external work.”

TOP 5: -40 C=F | Clinical 35-42 vs Thermocouple widest | Fusion 80 vs Vapor 539 | Water density max 4°C | C-C-R + Zeroth/First/Second law split.

PERIODIC & WAVE MOTION — MEMORY TRICK SHEET

General Science · Physics · Chapter 8 · SHM + Pendulum + Waves + EM Spectrum + Sound + Ultrasonic + Echo + Mach

Ye full notes nahi hai — source chapter ke SHM/pendulum relations, wave types, EM-spectrum discoverers, sound characteristics, hearing ranges, speed-of-sound effects, echo/reverberation aur Mach-number traps ko fast exam-recall format mein compress kiya gaya hai.

1. Periodic Motion vs SHM

Concept	Core Idea	Examples
Periodic Motion	Regular time interval ke baad repeat	Swing, fan, musical vibrations, satellite motion
SHM	Mean position ke about oscillation; acceleration mean ki taraf	Clock pendulum, spring-mass system

TRICK - “Periodic = Repeat; SHM = Restore to Mean”

SHM mein restoring acceleration hamesha mean position ki taraf hota hai.

EXAM TRAP -

SHM acceleration constant nahi hota; displacement ke saath vary karta hai.

2. SHM Formula Pack — T-n- ω

Quantity	Formula / Meaning
Time Period	$T = 2\pi/\omega$
Frequency	$n = 1/T$
Time Period-Frequency	$T = 1/n$
Amplitude	Maximum displacement from mean position
Phase	Direction/state of motion ko express karta hai

TRICK - “T and n = Reciprocal Twins”

$n = 1/T$. Higher frequency $\rightarrow$ shorter time period.

TRICK - “Amplitude = Maximum Displacement”

Mean position se maximum displacement ko amplitude bolte hain.

3. Simple Pendulum — $T = 2\pi\sqrt{l/g}$

- Simple pendulum = heavy point mass (bob) + light inextensible string + fixed support
- $T = 2\pi\sqrt{l/g}$
- T depends on length l and gravitational acceleration g
- T does NOT depend on bob mass

TRICK - “Pendulum T = Length UP, g DOWN”

l badhe $\rightarrow$ T badhe.

g ghate $\rightarrow$ T badhe.

EXAM TRAP - “Mass Changes? T Says NO.”

Bob mass ka time period par no effect.

4. Pendulum Conditions — Up/Down/Satellite/Summer/Moon

Condition	Effect on T	Reason
Altitude increases	T increases	g decreases
Depth increases	T increases	g decreases
Satellite	$T = \infty$	Effective $g = 0$ / weightlessness
Summer	T increases	Length increases
Winter	T decreases	Length decreases
Moon	T increases	$g_{\text{moon}} = g_{\text{earth}}/6$
Bob mass changes	No change	T independent of mass

TRICK - “UP bhi T ↑, DOWN bhi T ↑”

Earth surface se altitude aur depth dono directions mein g decreases, so T increases in source notes.

TRICK - “Satellite = Infinity; Moon = Longer”

Satellite mein pendulum oscillate nahi karta in source treatment.

EXAM TRAP - “Max Pendulum T = 84.6 min”

Source note: length equal to Earth radius ke liye maximum T = 84.6 minutes.

5. Wave Motion — $v = n\lambda$

- Wave = disturbance that transports energy from one place to another
- $v = \lambda/T = n\lambda$
- $\lambda = v/n$
- Frequency source-set property hai; medium change par unchanged rehti hai

TRICK - “Wave Triangle = $v - n - \lambda$ ”

$v = n\lambda$; $\lambda = v/n$.

EXAM TRAP -

Medium change par speed aur wavelength change ho sakte hain, frequency nahi.

6. Mechanical vs Electromagnetic Waves

Type	Particle Motion / Medium	Examples
Longitudinal Mechanical	Along wave direction; medium required	Sound, seismic waves, tsunamis, spring vibrations
Transverse Mechanical	Perpendicular; medium required	Transverse spring motion
Electromagnetic	Transverse; no material medium required	Light, radio, microwave, X-ray, gamma

TRICK - “Sound = Longitudinal + Mechanical + Medium”

Vacuum mein sound travel nahi karta.

TRICK - “EM = Transverse + Medium-Free”

Vacuum speed = 3×10^8 m/s.

EXAM TRAP -

Mechanical waves need medium; EM waves do not.

7. EM Spectrum Order — R-M-I-V-U-X-G

Wave	Discoverer (Year)	Key Use
Radio	Marconi (1895)	Radio & TV communication
Microwave	Heinrich Hertz (1888)	Satellites, wireless communication
Infrared	Herschel (1800)	Remote control, fog photography
Visible	Newton (1666)	Human vision, bulb
Ultraviolet	Ritter (1801)	Forged-document detection, food preservation
X-Ray	Roentgen (1895)	Bone fractures, lung diseases
Gamma	Paul Villard (1900)	Cancer treatment, nuclear structure

TRICK - “R-M-I-V-U-X-G”

Radio → Microwave → Infrared → Visible → UV → X-ray → Gamma.

Is order mein wavelength decreases, frequency increases.

TRICK - “Discoverers = M-H-H-N-R-R-V”

Marconi-Hertz-Herschel-Newton-Ritter-Roentgen-Villard.

EXAM TRAP - “Gamma Villard; Named Rutherford”

Microwaves generated by magnetron; visible $\approx$ 400 nm violet to 700 nm red.

8. Sound Characteristics — L-P-Q-I

Characteristic	Meaning / Relation	Exam Hook
Loudness	Intensity-linked; Loudness $\propto$ Amplitude ²	Also falls approx $1/\text{distance}^2$
Pitch	Shrillness; $\propto$ Frequency	Higher f = higher pitch
Quality / Timbre	Distinguishes same pitch & loudness	Tone one frequency; Note many
Intensity	Sound energy per area per second	Unit W/m ²

TRICK - “Loudness = Amplitude²; Pitch = Frequency”

L-A² / P-f.

EXAM TRAP - “Tone One; Note Many”

Single frequency vs multiple frequencies.

9. Sound Levels — Quiet to Jet

Source	Loudness
Normal breathing	10 dB
Whisper	15-30 dB
General conversation	30-60 dB
Busy traffic	70 dB
Jet plane	140-150 dB

TRICK - “10 → 30 → 60 → 70 → 150”

Breathing → whisper/conversation → traffic → jet.

EXAM TRAP - “85 dB+ Harmful”

Source flags noise above 85 dB as harmful to human ears.

10. Hearing Range — I-A-U

Range	Category	Examples / Use
< 20 Hz	Infrasonic	Seismic waves, heartbeat
20-20,000 Hz	Audible	Adult human hearing
> 20,000 Hz	Ultrasonic	Bat, dog, dolphin, cat; SONAR; medical imaging

TRICK - “I-A-U = Below20 / 20-20k / Above20k”

Infrasonic-Audible-Ultrasonic.

TRICK - “SONAR = Sound Navigation And Ranging”

Uses ultrasonic waves.

EXAM TRAP -

Adult hearing range = 20 Hz to 20 kHz.

11. Animal Hearing — D-B-D-C-H

Animal / Entity	Source Range
Dolphin	150-1,50,000 Hz
Bat	1,000-1,20,000 Hz
Dog	15-45,000 Hz
Cat	60-65,000 Hz
Children	Up to 25,000 Hz
Human Adult	20-20,000 Hz

TRICK - “Dolphin Highest Listed 150k; Bat 120k”

Source animal-range table ka fast recall.

12. Ultrasonic Applications — M-S-D-I

- Medical imaging: ultrasonography + echocardiography
- SONAR: sea depth measurement
- Detecting submarines and icebergs
- Industrial cleaning and welding

TRICK - “M-S-D-I = Medical - SONAR - Detection - Industrial”

Ultrasonic uses ka four-part chain.

13. Speed of Sound — Solid > Liquid > Gas

TRICK - “S-L-G = Fast to Slow”

Speed of sound: Solid > Liquid > Gas.

Medium	Speed at 25°C
Aluminium	6420 m/s
Steel	5960 m/s
Sea Water	1531 m/s
Distilled Water	1498 m/s
Hydrogen	1284 m/s
Air	340 m/s

TRICK - “Aluminium Fastest Listed; Air ≈ 340”

Source quick-number pair.

14. Temperature, Pressure, Humidity Effects

Condition	Effect on Sound Speed
Temperature increases	Speed increases; $v \propto \sqrt{T}$
Pressure changes at constant T	No effect
Humidity increases	Speed increases; moist air lighter

TRICK - “ $T \uparrow = v \uparrow$; Pressure@sameT = NO effect; Humidity $\uparrow = v \uparrow$ ”

T-P-H trend.

Temperature	Speed in Air
0°C	332 m/s
22°C	344 m/s
25°C	346 m/s

EXAM TRAP -

At constant temperature, pressure has NO effect on sound speed.

15. Echo vs Reverberation

Term	Meaning	Source Key
Echo	Distinct re-hearing of reflected sound	Persistence 0.1 s; min distance 17.2 m
Reverberation	Prolonged persistence by multiple reflections	Halls; megaphones/loudspeakers/stethoscope context

TRICK - “Echo = Repeat; Reverb = Remains”

Echo distinct repetition; reverberation prolonged persistence.

EXAM TRAP - “0.1 s → 17.2 m”

Persistence of hearing and source minimum echo distance.

16. Mach Number — Sub / Super / Hyper

- Mach Number = speed of body / speed of sound in same medium
- Named after Ernst Mach
- Sonic boom occurs when aircraft crosses sound barrier near Mach 1

Mach Range	Class	Example
Mach < 1	Subsonic	Commercial aircraft ≈ 0.8
$1 < \text{Mach} \leq 5$	Supersonic	Fighter jets
Mach > 5	Hypersonic	Space shuttle during re-entry

TRICK - “Sub <1; Super 1-5; Hyper >5”

Source classification.

EXAM TRAP -

Hypersonic = Mach > 5 in source notes.

17. Formula + Number Flash

Item	Recall
SHM time period	$T = 2\pi/\omega$
Pendulum	$T = 2\pi\sqrt{l/g}$
Frequency	$n = 1/T$
Wave speed	$v = n\lambda$
Wavelength	$\lambda = v/n$
Sound vs temperature	$v \propto \sqrt{T}$
EM speed vacuum	$3 \times 10^8 \text{ m/s}$
Adult hearing	20-20,000 Hz
Persistence of hearing	0.1 s
Minimum echo distance	17.2 m
Mach number	$v_{\text{body}} / v_{\text{sound}}$

18. Final Exam Traps — 10 Red Flags

- SHM acceleration always mean position ki taraf; constant nahi
- Pendulum T bob mass par depend nahi karta
- Altitude aur depth dono increase karne par T increases in source treatment
- Satellite pendulum $T = \infty$ in source notes
- Sound longitudinal mechanical; EM/light transverse
- Medium change par frequency unchanged
- Pressure at constant temperature has no effect on sound speed
- Gamma discovered by Villard but named by Rutherford
- SONAR uses ultrasonic waves
- Echo min distance = 17.2 m; Hypersonic = Mach > 5

EXAM TRAP - “SHM-Mass-UpDown-Satellite-Sound/EM-Frequency-Pressure-Gamma-SONAR-Echo/Mach”

Chapter ke highest-yield traps isi sequence mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Periodic repeats; SHM restores to mean, acceleration toward mean not constant → $T=2\pi/\omega$, $n=1/T$; pendulum $T=2\pi\sqrt{l/g}$, mass no effect; altitude ↑/depth ↑/summer/moon all $T \uparrow$, winter $T \downarrow$, satellite $T=\infty$, source max 84.6min → Wave $v=n\lambda$, medium change frequency same → Sound longitudinal mechanical medium-needed vs EM transverse medium-free speed 3×10^8 → EM order R-M-I-V-U-X-G, wavelength ↓ frequency ↑; discoverers Marconi-Hertz-Herschel-Newton-Ritter-Roentgen-Villard; gamma named Rutherford → Sound L-P-Q-I: Loudness $\propto A^2$, Pitch $\propto f$, Tone one/Note many; >85dB harmful → hearing I-A-U: <20 infra, 20-20k audible, >20k ultrasonic; SONAR full form + ultrasound → Sound speed $S > L > G$; Al6420, air ≈ 340 ; $T \uparrow v \uparrow$, pressure@sameT no effect, humidity $\uparrow v \uparrow$ → Echo distinct 0.1s / 17.2m vs Reverb prolonged → Mach = body/sound; Sub<1, Super1-5, Hyper>5; sonic boom near Mach1.”

TOP 5: Pendulum mass no effect | Sound longitudinal vs EM transverse | R-M-I-V-U-X-G + discoverers | Frequency unchanged across medium | Echo 17.2 m + Hypersonic > Mach 5.

ELECTROSTATICS — MEMORY TRICK SHEET

General Science · Physics · Chapter 9 · Charge + Coulomb + Field + Potential + Current + Resistance + Cells + Instruments

Ye full notes nahi hai — source chapter ke charge laws, electric field/potential, capacitance, current, Ohm law, resistance combinations, heating effect, cells aur measuring instruments ke high-confusion facts ko fast exam-recall format mein compress kiya gaya hai.

1. Charge — A-Q-C-R

Property	Rule
Additive	$Q = q_1 + q_2 + q_3$
Quantisation	$Q = ne$; $e = \pm 1.6 \times 10^{-19} \text{ C}$
Conservation	Isolated system ka total charge constant
Repel / Attract	Like repel; unlike attract

TRICK - “A-Q-C-R = Add - Quantise - Conserve - Repel/Attract”

Charge properties ka four-part code.

EXAM TRAP - “Same Repels; Opposite Attracts”

Electric force gravity ki tarah sirf attractive nahi — source notes attraction + repulsion dono batate hain.

2. Coulomb’s Law — $K q_1 q_2 / r^2$

- $F = K(q_1 q_2) / r^2$
- $K = 9 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$
- $K = 1 / (4\pi\epsilon_0)$
- $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 / (\text{N} \cdot \text{m}^2)$
- Inverse-square law; source treats point charges

TRICK - “Charge Product UP; Distance-Square DOWN”

$q_1 q_2$ numerator; r^2 denominator.

EXAM TRAP - “ $K = 9 \times 10^9$; $e = 1.6 \times 10^{-19}$ ”

Chapter ke two core constants.

3. Conductor - Insulator - Semiconductor

Class	Source Behaviour	Examples
Conductor	Free electrons; charge flows easily	Silver, Gold, Copper, Graphite, Sea water
Insulator	Charge does not flow easily	Rubber, Oil, Polythene, Pure water, Lead - as listed in source
Semiconductor	Between conductor and insulator	Silicon, Germanium

TRICK - “Silver Best; Pure Water Insulator; Sea Water Conductor”

Sea water dissolved salts/ions ki wajah se conduct karta hai.

EXAM TRAP -

Best conductor in source = Silver, not Copper.

4. Electric Field — P→N, Never Cross

Item	Recall
Electric field intensity	$E = F/q_0$; vector; N/C
Field-line direction	Positive → Negative
Intersection	Never intersect
Closed loops	Do not form closed loops
Strength	Closer lines = stronger field
Conductor surface	Lines perpendicular to surface

TRICK - “P-to-N; Never Cross; Perpendicular at Conductor”

Field-lines ka 3-rule pack.

EXAM TRAP -

Field lines intersect karein to ek point par do directions ho jaayengi — source explanation.

5. Potential + Capacitance — W/q and Q/V

Quantity	Formula	Nature / Unit
Electric Potential	$V = W/q_0$	Scalar; Volt = J/C
Potential Difference	$V_a - V_b = W/q_0$	Volt
Capacitance	$C = Q/V$	Farad = C/V

TRICK - “Potential = Work/Charge; Capacitance = Charge/Volt”

$V = W/q$; $C = Q/V$.

EXAM TRAP - “Farad, Not Volt/Ohm”

Practical units: μF and pF .

- Source solved potential example: $W=2.25 \text{ J}$, $q=12 \times 10^{-4} \text{ C} \rightarrow V=1875 \text{ V}$
- Source solved capacitor example: $C=1.0 \mu\text{F}$, $V=6.0 \text{ kV} \rightarrow Q=6 \times 10^{-3} \text{ C}$

6. Electric Current — $i = q/t$

- Current $i = q/t$
- SI unit Ampere = C/s
- $1 \text{ A} \approx 6.25 \times 10^{18}$ electrons per second
- Conventional current: high potential → low potential
- Electron flow direction conventional current ke opposite
- Solids: free electrons; liquids: ions + electrons; semiconductors: holes + electrons

TRICK - “Current Goes High→Low; Electrons Go Opposite”

Conventional current vs electron-flow direction.

EXAM TRAP -

Current direction aur electron direction same nahi.

7. AC vs DC

Type	Direction	Typical Use
AC	Periodically alternates	Household supply, generators, transformers
DC	Remains constant	Batteries, electronic circuits, charging

TRICK - “AC Alternates; DC Direct”

Naam hi direction rule yaad kara deta hai.

8. Ohm's Law — $V = IR$

- Proposed by George Simon Ohm in 1827
- $V \propto I$ when temperature and other physical conditions remain constant
- $V = IR$

Resistance Type	Source Recall
Ohmic	Obeys $V=IR$; V-I graph straight; source example manganese wire
Non-ohmic	Does not obey; V-I graph non-linear; diode, bulb

TRICK - "Ohm 1827 + Constant Conditions"

Temperature/physical conditions unchanged hona mandatory.

EXAM TRAP -

Ohm law har condition mein valid nahi — source specifically constant physical conditions kehta hai.

9. Resistance — Long More, Area More Less

- $R = \rho l/A$
- $R \propto l$
- $R \propto 1/A$
- Resistivity ρ material property; unit $\Omega \cdot m$

TRICK - "Length $\uparrow \rightarrow R \uparrow$; Area $\uparrow \rightarrow R \downarrow$ "

$R = \rho l/A$.

10. Series vs Parallel — Same I vs Same V

Feature	Series	Parallel
Equivalent R	$R_1 + R_2 + R_3$	$1/R_{eq} = 1/R_1 + 1/R_2 + 1/R_3$
Current	Same	Different branches
Voltage	Different	Same
Equivalent R	Greater than any individual	Less than smallest individual
Source Use	Fuse, Ammeter	Domestic wiring, Voltmeter

TRICK - "Series = Same I; Parallel = Same V"

Domestic wiring = parallel.

TRICK - "Series R Grows; Parallel R Shrinks"

Series equivalent > individual; parallel equivalent < smallest.

EXAM TRAP -

Domestic wiring series nahi, parallel.

11. ρ - G - σ

Quantity	Formula	Unit
Resistivity ρ	$\rho = RA/l$	$\Omega \cdot m$
Conductance G	$G = 1/R$	Siemens S or Ω^{-1}
Specific Conductivity σ	$\sigma = 1/\rho$	$\Omega^{-1} m^{-1}$

TRICK - "Resistance ka Reciprocal = Conductance; Resistivity ka Reciprocal = Conductivity"

$G = 1/R$; $\sigma = 1/\rho$.

TRICK - "Silver $1.60e-8$; Copper $1.62e-8$ "

Source resistivity table mein Silver slightly lower $\rightarrow$ best conductor.

12. Joule Heating + Power — H-W-P

Quantity	Formula
Heating	$H = I^2Rt$ joules = $I^2Rt/4.2$ calories
Electrical Work / Energy	$W = VIt = I^2Rt = V^2t/R$
Power	$P = VI = V^2/R = I^2R$

TRICK - “Heat I^2Rt ; Work adds t ; Power removes t ”

H/W/P formulas ko family ki tarah yaad karo.

EXAM TRAP - “1 kWh = 1 unit electricity”

Source high-frequency recall.

13. Bulb - Heater - Fuse - MCB

Device	Material / Principle	Key Detail
Bulb	Tungsten filament	High resistivity; melting point 3422°C; N ₂ + Argon
Electric Iron	Nichrome	Source: 80% Nickel + 20% Chromium
Heater	Ceramic plate + Nichrome	Resistance heating
Fuse	Low-melting alloy	Source: Cu + Sn + Pb; melts to break circuit
MCB	Automatic protective device	Trips and resettable

TRICK - “Bulb = Tungsten; Heater/Iron = Nichrome”

Do not swap.

EXAM TRAP - “Fuse Melts; MCB Trips + Resets”

Protection-device distinction.

14. Electrolysis — Faraday

- Current through acidic conducting solution causes ionic decomposition
- Source associates law of electrolysis with Faraday
- Applications: electric cell and electroplating

TRICK - “Faraday = Electrolysis; Cell + Plating”

Chemical effect of electric current.

15. Cell Basics — Primary vs Secondary

Type	Source Energy / Rechargeability
Primary	Chemical → Electrical; cannot be recharged
Secondary	Charging: Electrical → Chemical; Discharge: Chemical → Electrical; rechargeable

TRICK - “Primary = One-Time; Secondary = Recharge”

Rechargeability split.

- EMF $E = W/q$; unit Volt
- Terminal potential $V = E - ir$
- Cell current $I = E/(R+r)$
- Internal resistance source relation: $r = (E/V - 1)R$

16. Cell Voltage Memory

Cell	EMF
Voltaic	1.08 V
Leclanche	1.46 V
Dry Cell	1.46 V
Daniell	1.1 V
Bichromate	2.0 V
Weston Cadmium	1.018 V
Lead Accumulator	2.2 V
Alkaline Accumulator	1.35 V

TRICK - “Dry/Leclanche = 1.46; Lead = 2.2”

Source ke most important cell values.

EXAM TRAP -

Lead accumulator 2.2 V; dry cell 1.46 V.

17. Ammeter vs Voltmeter vs Galvanometer

Instrument	Measures	Connection	Ideal R
Ammeter	Current	Series	0
Voltmeter	Potential difference	Parallel	∞
Galvanometer	Current detect + direction + magnitude	—	Medium

TRICK - “A = Series + Zero R; V = Parallel + Infinite R”

Ammeter aur voltmeter ka complete exam pair.

EXAM TRAP -

Ammeter parallel nahi; voltmeter series nahi.

18. Galvanometer Conversion — Opposite Additions

Conversion	Add
Galvanometer → Ammeter	Low resistance shunt in parallel
Galvanometer → Voltmeter	High resistance in series

TRICK - “To A: Low Parallel; To V: High Series”

Low-P / High-S.

EXAM TRAP -

Galvanometer→Ammeter: low R parallel; →Voltmeter: high R series.

19. Final Exam Traps — 12 Red Flags

- Best conductor in source = Silver, not copper
- Pure water = insulator; sea water = conductor
- Field lines Positive→Negative and never intersect
- Potential scalar; capacitance unit Farad
- Conventional current opposite electron motion
- Ohm law only at constant temperature/physical conditions
- Series = same current; Parallel = same voltage; domestic wiring parallel
- Bulb = Tungsten; heater/iron = Nichrome
- Faraday = electrolysis in source
- Dry/Leclanche = 1.46 V; Lead accumulator = 2.2 V
- Ammeter series $R=0$; Voltmeter parallel $R=\infty$
- Galvanometer→A: low-R parallel; →V: high-R series

EXAM TRAP - “Silver-Water-Field-Farad-Current-Ohm-Series-Bulb-Faraday-Cells-A/V-Galvano”

12 exam traps isi chain mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Charge A-Q-C-R; $e=1.6\times10^{-19}\text{C}$; Coulomb $F=Kq_1q_2/r^2$, $K=9\times10^9$, $\epsilon_0=8.85\times10^{-12}$ → Silver best conductor; pure water insulator, sea water conductor; semiconductor Si-Ge → Field $E=F/q$ vector N/C, lines P→N, never cross/loop, conductor surface perpendicular → Potential $V=W/q$ scalar Volt= J/C ; Capacitance $C=Q/V$ Farad → Current $i=q/t$, $1\text{A}\approx6.25\times10^{18}$ e/s, conventional High→Low and electrons opposite; AC alternates, DC constant → Ohm $V=IR$ only constant conditions; $R=\rho l/A$: length ↑ R ↑, area ↑ R ↓ → Series same I and R grows; Parallel same V and R shrinks; domestic parallel → $\rho=RA/l$, $G=1/R$, $\sigma=1/\rho$; Joule $H=I^2Rt$, $W=VIt=I^2Rt=V^2t/R$, $P=VI=I^2R=V^2/R$; 1kWh=1 unit → Bulb tungsten; iron/heater nichrome; fuse melts, MCB trips/reset; Faraday electrolysis → Primary one-use vs Secondary rechargeable; $E=W/q$, $V=E-ir$, $I=E/(R+r)$; Dry/Leclanche 1.46V, Lead 2.2V → Ammeter series $R=0$; Voltmeter parallel $R=\infty$; Galvanometer to A low-R parallel, to V high-R series.”

TOP 5: Silver best + pure/sea water split | Field P→N never crosses | Series same I / Parallel same V | Tungsten vs Nichrome | Ammeter series $R=0$ vs Voltmeter parallel $R=\infty$.

MAGNETISM — MEMORY TRICK SHEET

General Science · Physics · Chapter 10 · Magnets + Materials + Magnetic Field + Lorentz Force + Fleming Rules + Flux + EMI

Ye full notes nahi hai — source chapter ke magnet types, magnetic-material classes, Oersted/Faraday timeline, field rules, Lorentz force, Fleming rules, magnetic flux aur electromagnetic induction ke high-memory facts ko fast exam recall format mein compress kiya gaya hai.

1. Magnet Basics — Dipole Always

- Magnet iron aur uske ores ko attract karta hai
- Every magnet has North + South pole; isolated magnetic monopole does not exist
- Like poles repel; unlike poles attract
- Freely suspended bar magnet ka North pole geographic North ki taraf point karta hai
- Magnet ko cut karne par har piece phir N-S dipole ban jaata hai

TRICK - “Magnet = N + S Forever”

Cut karo, phir bhi monopole nahi milta; smaller dipoles milte hain.

EXAM TRAP -

Magnetic monopole does NOT exist in source notes.

2. Natural vs Artificial Magnet

Type	Source Detail	Examples
Natural	Nature mein found; irregular; weaker	Lodestone / Magnetite = Fe_3O_4
Artificial	Man-made magnetic materials	Bar magnet, electromagnet, horse-shoe, compass

TRICK - “Lodestone = Magnetite = Fe_3O_4 ”

Natural magnet ka direct equation.

EXAM TRAP -

Fe_3O_4 yaad karo; $\text{FeO}/\text{Fe}_2\text{O}_3$ nahi.

3. Magnetic Materials — Dia / Para / Ferro

Class	Behaviour	Examples from Source
Diamagnetic	Opposite magnetisation; weakly repelled	Bismuth, Zinc, Gold, Copper, Silver, Diamond, Salt
Paramagnetic	Field direction; weakly attracted	Chromium, Sodium, Aluminium, Oxygen
Ferromagnetic	Domain formation; strongly attracted	Iron, Nickel, Cobalt, Steel

TRICK - “Dia = Door; Para = Passes Near; Ferro = Full Attraction”

Dia repels weakly; Para attracts weakly; Ferro attracts strongly.

TRICK - “Ferro = I-N-C-S”

Iron - Nickel - Cobalt - Steel.

EXAM TRAP -

Silver/Copper = Diamagnetic; Aluminium = Paramagnetic.

4. Oersted vs Faraday — 1820 vs 1831

Scientist	Year	Discovery
Oersted	1820	Current-carrying conductor produces magnetic field

Faraday	1831	Changing magnetic flux produces induced EMF/current
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TRICK - “Oersted: Current → Field; Faraday: Changing Flux → Current”

1820 → 1831 timeline.

EXAM TRAP -

Oersted aur Faraday ki discoveries interchange mat karna.

5. Biot-Savart Law — $I \cdot l \cdot \sin\theta / r^2$

- $B = (\mu_0/4\pi) \cdot (I\Delta l \sin\theta / r^2)$
- $B \propto I$
- $B \propto \Delta l$
- $B \propto \sin\theta$
- $B \propto 1/r^2$
- $\mu_0/4\pi = 10^{-7} \text{ N/A}^2$
- Source unit of $B = \text{N}/(\text{A} \cdot \text{m})$

TRICK - “Biot = $I \cdot L \cdot \sin / r^2$ ”

Current, length aur $\sin\theta$ numerator; distance-square denominator.

TRICK - “ $\mu_0/4\pi = 10^{-7}$ ”

Direct numerical recall.

6. Ampere’s Law

- $\oint B \cdot dl = \mu_0 I$
- Closed loop ke around magnetic-field line integral = $\mu_0 \times$ enclosed current
- Source use: symmetric current configurations such as solenoid and toroid

TRICK - “Ampere = Closed Loop + Enclosed Current”

Symmetry problems ka source hook.

7. Direction Rules — Right Hand + Cork Screw

Rule	Thumb / Motion	Fingers / Result
Right Hand Palm	Thumb → current I	Fingers → magnetic field B
Maxwell Right Hand	Thumb → current	Curled fingers → magnetic field
Cork Screw	Current upward	Field anti-clockwise
Cork Screw	Current downward	Field clockwise

TRICK - “Current UP = Anti-clockwise; DOWN = Clockwise”

Source cork-screw rule.

EXAM TRAP -

Up/anti-clockwise and down/clockwise ko ulta mat karna.

8. Lorentz Force — Moving Charge Only

- Moving charge in magnetic field: $F = qvB \sin\theta = q(\mathbf{v} \times \mathbf{B})$
- $F \propto q, v, B$
- $\theta = 0^\circ$ or $180^\circ \rightarrow F = 0$
- $\theta = 90^\circ \rightarrow F$ maximum
- Stationary charge $v=0 \rightarrow$ magnetic force zero

TRICK - “Lorentz = $q \cdot v \cdot B \cdot \sin\theta$ ”

Charge, speed, field aur angle.

TRICK - “Parallel ZERO; Perpendicular MAX”

$0^\circ/180^\circ \rightarrow 0$ force; $90^\circ \rightarrow \text{max}$.

EXAM TRAP -

Stationary charge in magnetic field experiences NO magnetic force.

9. Force on Current-Carrying Conductor

- $F = BIl \sin\theta$
- Force current aur magnetic field dono ke perpendicular act karta hai
- $\theta = 90^\circ \rightarrow$ force maximum
- Current parallel to B ($\theta=0^\circ$) $\rightarrow$ force zero

TRICK - “Conductor Force = $B \cdot I \cdot l \cdot \sin\theta$ ”

Lorentz qvB aur conductor BIl ko separate yaad karo.

10. Fleming Rules — Left Motor, Right Generator

Rule	Used For	Thumb	Forefinger	Middle Finger
LEFT	Motor / force	Force / motion	Magnetic field B	Current I
RIGHT	Generator / induced current	Motion of conductor	Magnetic field B	Induced current

TRICK - “LEFT = MOTOR; RIGHT = GENERATOR”

Both rules mein Forefinger = Field, Middle finger = Current.

TRICK - “F-B-I = Thumb-Fore-Middle on Left”

Left hand: Force - B - I.

EXAM TRAP -

Motor = Left; Generator/Dynamo = Right.

11. Magnetic Field Lines

Property	Source Recall
Outside magnet	North $\rightarrow$ South
Shape	Closed continuous loops
Intersection	Never intersect
Uniform field	Parallel + equally spaced
Neutral point	Magnetic field = 0
Strength	Closer lines = stronger field

TRICK - “N $\rightarrow$ S Outside + Closed + Never Cross”

Magnetic-field lines ka 3-rule pack.

12. Magnetic vs Electric Field Lines

Feature	Electric	Magnetic
Direction	$+$ $\rightarrow$ $-$	N $\rightarrow$ S outside magnet
Closed loops?	No	Yes
Intersect?	No	No
Start/end	On charges	No start/end; continuous

TRICK - “Electric Open; Magnetic Closed”

Dono non-intersecting.

EXAM TRAP -

Magnetic field lines closed loops form karte hain; electric field lines nahi.

13. Electromagnet — Soft Iron + Current

- Wire coil around iron core + current → electromagnet
- Core material = soft iron; easily magnetised and demagnetised
- More turns + more current + better core material → greater strength
- Magnetism only while current flows
- Applications: electric bells, scrap-iron cranes, MRI machines, electric motors

TRICK - “Soft Iron + Turns + Current = Strong Electromagnet”

Current off → magnetism off.

14. Magnetic Flux — $\Phi = BA \cos\theta$

Item	Recall
Formula	$\Phi = B \cdot A \cos\theta$
Angle θ	Between B and surface normal
SI unit	Weber Wb
Other SI expression	N·m/A
CGS unit	Maxwell
Conversion	1 Wb = 10^8 Maxwell

TRICK - “Flux = BA cos NORMAL-angle”

θ surface ke saath nahi, surface NORMAL ke saath measured hai.

EXAM TRAP - “Perpendicular to Surface = Flux MAX; Parallel to Surface = ZERO”

Because B aligns with normal at max flux.

TRICK - “1 Weber = 10^8 Maxwell”

10^8 direct conversion.

15. Electromagnetic Induction — Changing Flux Needed

- EMI = changing magnetic flux se conductor mein EMF + induced current generate hona
- Faraday discovered EMI in 1831
- Static magnet motionless near coil → no induced EMF
- Condition = magnetic flux must change

TRICK - “EMI = Change Flux, Not Just Field”

Static flux → no induction.

EXAM TRAP -

Presence of magnetic field alone is insufficient; flux must CHANGE.

16. EMI Applications

Application	Role
Dynamo / Generator	Mechanical → Electrical
Transformer	Changes AC voltage levels
Artificial Pacemaker	Regulates heartbeat
Other source examples	Induction cooktop; wireless charging

TRICK - “G-T-P-IW = Generator-Transformer-Pacemaker-Induction/Wireless”

EMI application chain.

EXAM TRAP - “Generator = EMI; Motor = Magnetic Force”

Generator/right hand vs Motor/left hand.

17. Formula + Number Flash

Item	Recall
Biot-Savart	$B = (\mu_0/4\pi) I \Delta l \sin\theta / r^2$
Ampere	$\oint B \cdot dl = \mu_0 I$
Lorentz Force	$F = qvB \sin\theta$
Conductor Force	$F = BIl \sin\theta$
Flux	$\Phi = BA \cos\theta$
$\mu_0/4\pi$	10^{-7} N/A^2
Oersted	1820
Faraday	1831
Flux Conversion	1 Wb = 10^8 Maxwell
Natural Magnet	Lodestone = Magnetite = Fe ₃ O ₄

18. Final Exam Traps — 10 Red Flags

- Lodestone = Magnetite = Fe₃O₄
- Magnetic monopoles do not exist
- Silver/Copper diamagnetic; Aluminium paramagnetic
- Oersted 1820 = current→field; Faraday 1831 = changing flux→EMI
- Stationary charge has zero Lorentz magnetic force
- Left hand = motor; Right hand = generator
- Magnetic field lines closed; electric field lines not closed
- 1 Weber = 10^8 Maxwell
- Flux angle is measured with surface normal
- EMI requires changing magnetic flux

EXAM TRAP - “Fe₃O₄-Monopole-Materials-O/F-Stationary-L/R-Lines-Wb-Normal-Change”

Chapter ke 10 exam traps isi order mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Magnet always N-S dipole; like repel/unlike attract; cut→smaller dipoles; Lodestone=Magnetite=Fe₃O₄ → Materials: Dia weak repel (Bi,Zn,Au,Cu,Ag,Diamond,Salt), Para weak attract (Cr,Na,Al,O₂), Ferro strong/domain (Fe,Ni,Co,Steel) → Oersted1820 current→magnetic field; Faraday1831 changing flux→EMI → Biot $B=(\mu_0/4\pi)I\Delta l \sin\theta/r^2$ with $\mu_0/4\pi=10^{-7}$; Ampere $\oint Bdl=\mu_0 I$ → Direction: right-hand thumb current/fingers field; current UP anti-clockwise, DOWN clockwise → Lorentz $F=qvB\sin\theta$: stationary0, parallel0, perpendicularmax; conductor $F=BIl \sin\theta$ → Fleming LEFT motor F-B-I, RIGHT generator motion-B-current → Magnetic lines N→S outside, closed, never intersect; electric open, magnetic closed → Electromagnet soft iron + turns/current strength; current off magnetism off → Flux $\Phi=BA\cos\theta$, θ with normal, Wb; 1Wb= 10^8 Maxwell; perpendicular-surface max, parallel-surface zero → EMI requires changing flux; generator, transformer, pacemaker, induction cooktop, wireless charging; generator=EMI, motor=magnetic-force.”

TOP 5: Fe₃O₄ Lodestone | Dia/Para/Ferro examples | Oersted1820 vs Faraday1831 | Left Motor vs Right Generator | Flux angle with NORMAL + 1 Wb= 10^8 Maxwell.

LIGHT — MEMORY TRICK SHEET

General Science · Physics · Chapter 11 · Reflection + Refraction + TIR + Mirrors + Lenses + Dispersion + Eye + Wave Optics + Photoelectric Effect

Ye full notes nahi hai — source chapter ke mirror/lens sign traps, TIR conditions, VIBGYOR, human-eye defects, optical instruments, wave-optics scientists aur photon/photoelectric-effect facts ko fast exam-recall format mein compress kiya gaya hai.

1. Light Basics — EM + Dual Nature

- Light = energy form; normally straight line mein travel karta hai
- Material medium required nahi hota
- Transverse electromagnetic wave
- Dual nature: wave + particle behaviour
- Vacuum speed = 3×10^8 m/s
- Visible wavelength = 3900-7800 Å

TRICK - “Light = No Medium + Transverse EM + Dual”

Vacuum speed 3×10^8 m/s; visible 3900-7800 Å.

Theory	Scientist / Year	Source Hook
Particle	Descartes 1637; Newton 1700	Reflection/refraction + particle behaviour
Wave	Huygens 1678	Interference, diffraction, polarisation

TRICK - “D-N Particle; H Wave”

Descartes/Newton = particle side; Huygens = wave side.

2. Reflection — i = r + Coplanarity

- Reflection = light ka original medium mein return after striking surface
- Law 1: angle of incidence = angle of reflection
- Law 2: incident ray, reflected ray and normal same plane mein

TRICK - “Reflection = i = r + Same Plane”

Angles normal se measured hote hain.

3. Plane Mirror — V-E-S-L

Property	Recall
Nature	Virtual
Orientation	Erect
Size	Equal to object
Special	Laterally inverted
Distance	Image behind = object distance in front

TRICK - “Plane Mirror = V-E-S-L”

Virtual - Erect - Same size - Lateral inversion.

EXAM TRAP -

Plane-mirror image real nahi hota.

4. Spherical Mirrors — Concave vs Convex

Mirror	Nature	Major Uses
Concave	Converging	Shaving, dentist, torch/headlight, solar cooker
Convex	Diverging	Rear-view mirror, eyeglass, street lights

TRICK - “Rear-view = Convex; Shaving/Dentist = Concave”

Convex wide field + diminished virtual image.

Concave close object F-P ke beech enlarged virtual image.

EXAM TRAP - “Convex Always V-E-D”

Virtual - Erect - Diminished.

5. Concave Mirror Image Chain

Object Position	Image Position	Nature
Infinity	F	Real, inverted, highly diminished
Beyond C	Between C & F	Real, inverted, diminished
At C	At C	Real, inverted, same size
Between C & F	Beyond C	Real, inverted, enlarged
At F	Infinity	Real, inverted, highly enlarged
Between F & P	Behind mirror	Virtual, erect, enlarged

EXAM TRAP - “Concave Virtual ONLY F-P”

Object between F and P → behind mirror, virtual erect enlarged.

6. Mirror Formula — PLUS

Formula / Sign	Recall
Mirror formula	$1/v + 1/u = 1/f$
Focal-radius	$f = R/2$
Magnification	$m = -v/u$
Convex f	Positive
Concave f	Negative
Object distance u	Negative in source sign convention

TRICK - “MIRROR = PLUS”

$$1/v + 1/u = 1/f.$$

TRICK - “f = R/2”

Source example: $f=10\text{ m} \rightarrow R=20\text{ m}$.

7. Refraction — Snell + Frequency Same

- Refraction = transparent medium change par light deviation
- Denser medium mein speed decreases and ray normal ki taraf bends
- Frequency remains unchanged
- Snell: $\sin i / \sin r = \text{constant} = \mu$
- $\mu = c/v$

TRICK - “Denser = Slower + Toward Normal; Frequency SAME”

Higher refractive index = optically denser = lower light speed.

EXAM TRAP -

Medium change par frequency unchanged.

8. Total Internal Reflection — D-R + $i > c$

- Condition 1: light denser → rarer medium
- Condition 2: incidence angle > critical angle
- $\sin c = 1/\mu$
- Applications: optical fibre, medical endoscope, diamond brilliance, mirage, shining air bubble/crack

TRICK - “TIR = Denser→Rarer + $i > c$ Critical”

D-R + $i > c$.

TRICK - “Diamond = Small Critical Angle $\approx 24^\circ$ ”

High refractive index → TIR easily.

EXAM TRAP -

TIR rarer→denser travel mein nahi hota.

9. Lenses — Convex vs Concave

Lens	Shape/Behaviour	Sign / Uses
Convex	Thicker centre; converging	Positive f ; farsightedness, magnifying glass, camera
Concave	Thinner centre; diverging	Negative f ; myopia

TRICK - “Convex = Converge = Positive; Concave = Cave-in = Diverge = Negative”

Power sign bhi same: convex +, concave -.

10. Lens Formula — MINUS

Formula	Recall
Lens formula	$1/v - 1/u = 1/f$
Magnification	$m = v/u = I/O$
Power	$P = 1/f$ (metre)
Unit	Diopter D

TRICK - “MIRROR PLUS, LENS MINUS”

Mirror: $1/v + 1/u = 1/f$.

Lens: $1/v - 1/u = 1/f$.

EXAM TRAP - “ $f = -20\text{ cm} \rightarrow P = -5\text{ D}$ ”

Negative power = concave lens.

11. Dispersion — VIBGYOR

Colour	Source Exam Point
Violet	Most deviated; most scattered
Blue	Strongly scattered; sky appears blue
Green	Primary colour of light
Red	Least deviated; least scattered

TRICK - “Violet MOST; Red LEAST”

Deviation aur scattering dono mein source same extreme pair deta hai.

TRICK - “VIBGYOR: Violet→Red = Deviation/Scattering Down”

Violet end high, red end low.

12. Rainbow + Colour Mixing

- Rainbow: refraction + TIR + refraction in water droplets
- Sun observer ke behind hota hai
- Primary colours of light = Red, Green, Blue
- Primary colours of pigment/paint = Red, Yellow, Blue

Mix	Result
Green + Red	Yellow
Blue + Red	Magenta
Blue + Green	Cyan
RGB	White

TRICK - “RGB = Light = White”

R+G=Yellow; B+R=Magenta; B+G=Cyan.

TRICK - “Complementary: R-C, G-M, B-Y”

Red-Cyan; Green-Magenta; Blue-Yellow.

13. Rayleigh Scattering — Shorter More

Observation	Reason
Violet scatters most	Shortest visible wavelength side
Sky appears blue	Blue strongly scattered + eye more sensitive to blue than violet
Sunrise/sunset red	Long path scatters shorter wavelengths away
Danger signal red	Red scatters least

TRICK - “Shorter $\lambda \rightarrow$ More Scatter; Red = Least”

Danger signal red because red scatters least.

EXAM TRAP -

Violet scatters more than blue, but sky appears blue in source explanation.

14. Human Eye — C-I-P-L-R-F-B

Part	Function / Hook
Cornea	Transparent front; light entry
Iris	Coloured opaque part; controls pupil size
Pupil	Controls amount of entering light
Eye lens	Adjustable converging lens
Retina	Real, inverted image forms
Fovea / Yellow spot	Sharpest vision
Blind spot	Zero light sensitivity
Ciliary muscles	Change lens focal length

TRICK - “Retina = Real + Inverted; Fovea = Finest; Blind = Zero”

Three eye-image traps.

- Least distance of distinct vision = 25 cm
- Normal vision range = 25 cm to infinity
- Power of human eye = 4 D as given in source

EXAM TRAP -

Retinal image real and inverted; brain interprets it upright.

15. Vision Defects — M-H-P-A

Defect	Source Symptom	Correction
Myopia	Near clear; far blurred	Concave lens
Hypermetropia	Far clear; near blurred	Convex lens
Presbyopia	Old age; near/far difficulty	Bifocal lens
Astigmatism	Blurred all distances; irregular curvature	Cylindrical lens
Colour blindness	Red-green distinction problem; hereditary	No cure / no lens correction

TRICK - “M-H-P-A = Concave-Convex-Bifocal-Cylindrical”

Myopia→Concave; Hypermetropia→Convex; Presbyopia→Bifocal; Astigmatism→Cylindrical.

EXAM TRAP -

Colour blindness has no lens correction in source.

16. Optical Instruments — Microscope vs Telescope

Instrument	Source Formula / Hook
Simple Microscope	Convex short-f lens; $M = 1 + D/F$, $D=25\text{ cm}$
Reflecting Telescope	Concave mirror objective; bright magnified distant images
Astronomical Telescope at ∞	$M = -F_o/F_e$
Telescope Setup	Objective large F_o ; eyepiece small F_e ; larger $F_o/F_e \rightarrow$ more magnification

TRICK - “Telescope: Objective LONG, Eyepiece SHORT”

F_o large; F_e small.

17. Wave Optics — Young / Diffraction / Polarisation

Phenomenon	Source Recall
Interference	Thomas Young 1801; bright constructive, dark destructive
Diffraction	Bending at edges/obstacles; Fresnel nearby source, Fraunhofer parallel rays
Polarisation	Vibrations restricted to one direction; proves light transverse

TRICK - “Young = Interference 1801”

Soap-bubble colours = common interference example in source.

EXAM TRAP - “Polarisation PROVES Transverse”

Not longitudinal.

18. Photoelectric Effect + Photon Theory

Scientist / Year	Contribution
Planck 1900	Quantum/photon theory; $E = h\nu$
Einstein 1905	Photoelectric effect explanation
Einstein Nobel 1921	For photoelectric effect

- Photoelectric effect: sufficient-frequency light falls on metal $\rightarrow$ electrons emitted
- Threshold frequency = minimum frequency needed
- Kinetic energy of emitted electrons depends on frequency, not intensity — source exam point
- Photon energy $E = h\nu$
- Photon momentum $p = h/\lambda$
- $v = c/\lambda$

TRICK - “Planck 1900 Photon; Einstein 1905 Photoelectric; Nobel 1921”

P-E-N timeline.

EXAM TRAP -

Einstein’s 1921 Nobel Prize source says photoelectric effect, not relativity.

19. Formula Flash

Topic	Formula / Recall
Mirror	$1/v + 1/u = 1/f$
Lens	$1/v - 1/u = 1/f$
Mirror magnification	$m = -v/u$
Lens magnification	$m = v/u$
Power	$P = 1/f(m) D$
Refractive index	$\mu = c/v$
Critical angle	$\sin c = 1/\mu$
Photon energy	$E = h\nu$
Photon momentum	$p = h/\lambda$
Frequency-wavelength	$v = c/\lambda$

20. Final Exam Traps — 12 Red Flags

- Mirror formula PLUS; lens formula MINUS
- Rear-view = convex; shaving/dentist = concave
- Convex mirror always virtual, erect, diminished
- Concave mirror virtual image only when object lies between F and P
- TIR requires denser→rarer and incidence angle > critical
- Convex lens power positive; concave lens power negative
- Violet most deviated/scattered; red least
- Sky appears blue despite violet scattering more
- Myopia concave; hypermetropia convex; presbyopia bifocal; astigmatism cylindrical
- Retinal image = real + inverted
- Polarisation proves light is transverse
- Planck1900 photon vs Einstein1905 photoelectric; Nobel1921 photoelectric

EXAM TRAP - “+/- - Mirrors - TIR - Power - V/R - Sky - Eye Defects - Retina - Polarisation - Planck/Einstein”

Chapter ke 12 highest-yield traps isi chain mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Light no-medium transverse EM + dual; $c=3\times10^8$, visible3900-7800Å; Descartes/Newton particle, Huygens wave → Reflection $i=r$ +same plane; plane mirror V-E-S-L → concave/convex: rear-view convex, shaving/dentist concave; convex always V-E-D; concave virtual only F-P → Mirror PLUS $1/v+1/u=1/f$, $f=R/2$, $m=-v/u$ → Refraction denser=slower+toward normal, frequency same; Snell $\sin i/\sin r=\mu=c/v$ → TIR denser→rarer + $i>c$, $\sin c=1/\mu$; fibre/endoscope/diamond/mirage → Convex lens + converge; concave - diverge; Lens MINUS $1/v-1/u=1/f$, $P=1/f D$ → Dispersion VIBGYOR: violet most, red least; rainbow refraction+TIR+refraction; RGB light→white; R-C/G-M/B-Y complementary → Rayleigh shorter scatters more; sky blue, sunset/danger red → Eye retina real-inverted, fovea sharp, blind spot zero, near point25cm; defects M-H-P-A = concave-convex-bifocal-cylindrical → Microscope $M=1+D/F$; telescope objective F_o large, eyepiece F_e small → Young1801 interference; diffraction edge-bending; polarisation proves transverse → Planck1900 photon $E=h\nu$, $p=h/\lambda$; Einstein1905 photoelectric, Nobel1921 for photoelectric.”

TOP 5: Mirror PLUS vs Lens MINUS | Rear-view convex / Shaving concave | TIR D→R + $i>c$ | M-H-P-A corrections | Planck1900 vs Einstein1905/Nobel1921.

SEMICONDUCTOR — MEMORY TRICK SHEET

General Science · Physics · Chapter 12 · Intrinsic + Extrinsic + Doping + n-Type/p-Type + Photodiode + Zener + LED + Solar Cell

Ye full notes nahi hai — source chapter ke intrinsic/extrinsic semiconductors, doping, n-Type/p-Type carrier traps, temperature behaviour aur semiconductor devices/principles ko fast exam-recall tricks mein compress kiya gaya hai.

1. Conductor - Semiconductor - Insulator

Material	Conductivity	Band Gap	Temperature Effect	Examples
Conductor	High	Zero / overlapping	Resistance increases	Ag, Cu, Au, Al
Semiconductor	Medium	Small (~1 eV)	Resistance decreases	Si, Ge
Insulator	Very low	Large (>5 eV)	Minimal change	Glass, Rubber, Diamond

TRICK - “C-S-I = High-Medium-Low Conductivity”

Conductor > Semiconductor > Insulator.

TRICK - “Semiconductor Heats → Resistance Drops”

Temperature ↑ → semiconductor conductivity ↑, resistance ↓.

Source contrasts this with conductor, where conductivity decreases / resistance rises.

EXAM TRAP -

Diamond = insulator, semiconductor nahi.

2. Intrinsic vs Extrinsic — Pure vs Doped

Feature	Intrinsic	Extrinsic
Purity	Pure Si or Ge	Doped Si or Ge
Impurity	None	Added intentionally
Carrier balance	Electrons = holes	One carrier becomes majority
Conductivity	Relatively low	Higher
Main types	—	n-Type, p-Type

TRICK - “Intrinsic = In-Pure; Extrinsic = Extra Impurity”

Intrinsic = pure semiconductor.

Extrinsic = doped semiconductor.

TRICK - “Si-Ge = Pure Pair”

Silicon + Germanium = intrinsic examples.

EXAM TRAP - “0 K Intrinsic = Perfect Insulator”

At absolute zero free carriers absent.

3. Doping — Small Impurity, Big Conductivity

- Doping = pure Si/Ge mein very small impurity add karna
- Purpose = available charge carriers increase karna → conductivity increase
- Added impurity = dopant
- Extrinsic semiconductor doping ke baad banta hai

TRICK - “Doping = Dopant Added → Carriers Added”

Conductivity increase is the exam core.

4. n-Type — Penta Gives Extra Electron

Point	n-Type
Dopant valency	Pentavalent = 5
Common dopants	Arsenic (As), Antimony (Sb), Phosphorus (P)
Bond idea	4 electrons bond; 5th becomes extra free electron
Majority carrier	Electrons (-)
Minority carrier	Holes
Meaning of n	Negative carriers = electrons

TRICK - “n = 5 → Extra Electron”

Pentavalent dopant gives one extra electron.

TRICK - “A-S-P = n-Type”

Arsenic - Antimony - Phosphorus.

EXAM TRAP -

Phosphorus doping gives n-Type.

5. p-Type — Tri Leaves One Hole

Point	p-Type
Dopant valency	Trivalent = 3
Common dopants	Aluminium (Al), Boron (B), Gallium (Ga), Indium (In)
Bond idea	One electron missing → hole
Majority carrier	Holes (+)
Minority carrier	Electrons
Meaning of p	Positive carriers = holes

TRICK - “p = 3 → One Hole”

Trivalent dopant leaves one bond incomplete.

TRICK - “A-B-G-I = p-Type”

Aluminium - Boron - Gallium - Indium.

EXAM TRAP -

Aluminium gives p-Type even though aluminium itself is a good conductor.

6. n-Type vs p-Type — 5E vs 3H

Property	n-Type	p-Type
Dopant	Pentavalent (5)	Trivalent (3)
Examples	As, Sb, P	Al, B, Ga, In
Majority carrier	Electrons	Holes
Minority carrier	Holes	Electrons
Carrier sign	Negative	Positive
Overall charge	Neutral	Neutral

TRICK - “5 → Electron; 3 → Hole”

n = pentavalent + electrons.

p = trivalent + holes.

EXAM TRAP - “Carrier Sign $\neq$ Net Charge”

Both n-Type and p-Type electrically neutral overall.

7. Photodiode — Light Detection

Item	Photodiode
Purpose	Optical / light detection
Conversion	Light $\rightarrow$ electrical signal
Principle	Photoelectric effect
Use	Light intensity detection; light-operated keys

TRICK - “Photodiode = PHOTOelectric = Detect Light”

Photoelectric effect.

8. Zener Diode — Reverse Bias Regulator

Item	Zener Diode
Main purpose	Voltage regulation / voltage control
Operation	Reverse bias
Region	Reverse-breakdown region
Result	Nearly constant load voltage

TRICK - “Zener = Zero-Worry Voltage in Reverse”

Reverse-bias breakdown $\rightarrow$ voltage regulation.

EXAM TRAP -

Zener ko ordinary forward-biased diode ki tarah treat mat karna.

9. LED — Electrical to Light

Item	LED
Full form	Light Emitting Diode
Conversion	Electrical energy $\rightarrow$ radiant/light energy
Principle	Electroluminescence
Inventor	Nick Holonyak
Uses	Remote control, burglar alarm, optical communication

TRICK - “LED = Electricity $\rightarrow$ Light = Electroluminescence”

Inventor = Nick Holonyak.

EXAM TRAP -

LED principle = Electroluminescence; photoelectric/photovoltaic nahi.

10. Solar Cell — Light to Electrical

Item	Solar Cell
Conversion	Solar/light energy $\rightarrow$ electrical energy
Principle	Photovoltaic effect
Uses	Remote-area power generation; artificial-satellite batteries

TRICK - “Solar = PhotoVOLTAlC = Light $\rightarrow$ Voltage/Electricity”

Photovoltaic effect.

11. Device Principle Master — P-Z-L-S

Device	Energy / Function	Principle
Photodiode	Light → electrical signal	Photoelectric effect
Zener diode	Voltage regulation	Reverse breakdown
LED	Electrical → light	Electroluminescence
Solar cell	Light → electrical	Photovoltaic effect

TRICK - “PhotoDIODE = Photoelectric; Solar = Photovoltaic; LED = Electroluminescence”

Teen similar-looking principles ko alag-alag lock karo.

EXAM TRAP - “Zener = Reverse Breakdown”

Function-based fourth device.

12. Temperature Behaviour — Semiconductor vs Conductor

Material	Temperature ↑	Result
Semiconductor	Conductivity ↑	Resistance ↓
Conductor	Conductivity ↓	Resistance ↑

TRICK - “Semi Hot = Conduct More; Metal Hot = Resist More”

Opposite temperature behaviour.

EXAM TRAP -

Semiconductor conductivity temperature ke saath increases.

13. Rapid-Fire Recall

Question	Answer
Key intrinsic semiconductors?	Silicon, Germanium
Doping means?	Adding impurity
Added impurity called?	Dopant
n-Type valency?	Pentavalent 5
n-Type majority?	Electrons
p-Type valency?	Trivalent 3
p-Type majority?	Holes
Aluminium doping?	p-Type
Phosphorus doping?	n-Type
Intrinsic at 0 K?	Perfect insulator
Photodiode principle?	Photoelectric effect
Zener use?	Voltage regulation
LED principle?	Electroluminescence
LED inventor?	Nick Holonyak
Solar-cell principle?	Photovoltaic effect

14. Final Exam Traps — 10 Red Flags

- Semiconductor resistance decreases as temperature increases
- Silicon and Germanium = intrinsic semiconductor examples
- At 0 K intrinsic semiconductor = perfect insulator
- n-Type = pentavalent 5; electrons majority
- p-Type = trivalent 3; holes majority
- Aluminium gives p-Type; Phosphorus gives n-Type
- n-Type/p-Type overall electrically neutral
- Photodiode = photoelectric effect
- LED = electroluminescence; inventor Nick Holonyak
- Solar cell = photovoltaic; Zener = reverse-bias voltage regulation

EXAM TRAP - “Temp-SiGe-0K-5E-3H-Al/P-Neutral-PhotoD-LED-Solar/Zener”

Chapter ke 10 highest-yield traps isi order mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“C-S-I = high-medium-low conductivity; semiconductor small $\sim 1\text{eV}$ band gap, Si-Ge examples, heat $\uparrow$ conductivity $\uparrow$ resistance $\downarrow$; diamond insulator $\rightarrow$ Intrinsic = pure Si/Ge, e=holes, low conductivity, 0K perfect insulator; Extrinsic = doped, higher conductivity, n/p types $\rightarrow$ Doping = tiny impurity add $\rightarrow$ carriers $\uparrow \rightarrow$ conductivity $\uparrow \rightarrow$ n-Type: PENTA5, A-S-P = As-Sb-P, one extra electron, electrons majority, holes minority $\rightarrow$ p-Type: TRI3, A-B-G-I = Al-B-Ga-In, one hole, holes majority, electrons minority $\rightarrow$ both overall neutral $\rightarrow$ Photodiode light $\rightarrow$ electrical by photoelectric; Zener reverse-breakdown voltage regulator; LED electrical $\rightarrow$ light by electroluminescence, Nick Holonyak; Solar light $\rightarrow$ electrical by photovoltaic $\rightarrow$ Master split: PhotoD=Photoelectric, Solar=Photovoltaic, LED=Electroluminescence, Zener=Reverse Breakdown.”

TOP 5: Heat $\uparrow \rightarrow$ semiconductor R $\downarrow$ | n-Type 5/electron vs p-Type 3/hole | Both neutral | Photodiode-photoelectric vs Solar-photovoltaic vs LED-electroluminescence | Zener reverse-bias regulator.

OUR UNIVERSE — MEMORY TRICK SHEET

General Science · Physics · Chapter 13 · Big Bang + Galaxies + Stars + Sun + Solar System + India & Space

Ye full notes nahi hai — source chapter ke universe theories, red/blue shift, galaxy types, star facts, Sun, planets aur Indian space missions ke high-memory names, years, numbers aur exam traps ko fast recall format mein compress kiya gaya hai.

1. Universe — Number Hooks

Fact	Source Answer
Age of Universe	13.8 billion years
Big Bang	George Lemaitre — 1930
Hubble constant	67 km/s/Mpc
Total stars	$\approx 10^{22}$
Our galaxy	Milky Way — Spiral
Nearest star after Sun	Proxima Centauri
Brightest night-sky star	Sirius — Dog Star
Sunlight to Earth	8 min 20 sec
Chandrasekhar limit	1.44 solar masses

TRICK - “13.8 - 67 - 10^{22} - 8:20 - 1.44”

Universe age → Hubble constant → stars → sunlight time → Chandrasekhar limit.

2. Universe Theories — P-C-H-H-L

Theory	Propounder	Year / Core
Geocentric	Ptolemy	Ancient; Earth at centre
Heliocentric	Copernicus	1543; Sun at centre
Herschel Theory	Herschel	1785; Milky Way shape/structure
Hubble Theory	Edwin Hubble	1925; expanding universe
Big Bang	George Lemaitre	1930; origin ~13.8 bya

TRICK - “P-C-H-H-L = Earth-Sun-Galaxy-Expansion-Origin”

Ptolemy Earth-centre; Copernicus Sun-centre; Herschel Milky Way; Hubble expansion; Lemaitre Big Bang.

EXAM TRAP - “Big Bang = Lemaitre; Expansion = Hubble”

Ye do facts alag-alag hain.

3. Red Shift - Blue Shift - Hubble

Concept	Meaning	Exam Interpretation
Red Shift	Longer/red wavelength	Moving away
Blue Shift	Shorter/blue wavelength	Coming closer
Hubble Law	$V = Hd$	Recession velocity = $H \times \text{distance}$

TRICK - “RED = Recedes; BLUE = Back Toward Us”

Most distant galaxies red shift; source gives Andromeda as blue-shift example.

EXAM TRAP - “ $V = Hd$; $H = 67$ ”

Source Hubble constant = 67 km/s/Mpc.

4. Galaxies — E-S-I

Type	Shape	Stars/Colour	Source Example
Elliptical	Oval	Old; yellow-red; minimum formation	Maffei-1
Spiral	Spiral arms	Arms young/blue; centre old/red; maximum formation	Milky Way
Irregular	No regular shape	Mixed; variable; intermediate formation	Sextans A

TRICK - “E-S-I = Maffei - Milky - Sextans”

Elliptical Maffei-1; Spiral Milky Way; Irregular Sextans A.

EXAM TRAP - “Spiral Arms Blue; Centre Red”

Young hot stars in arms; older stars in central bulge.

5. Star Facts — P-S-Pole-Sun

Cue	Answer
Nearest star after Sun	Proxima Centauri
Brightest night-sky star	Sirius — Dog Star
Pole Star	Northern direction; navigation
Sun	Yellow star of Milky Way

EXAM TRAP - “Proxima = Nearest; Sirius = Brightest”

Nearest aur brightest ko interchange mat karna.

6. Star Composition + Colour Temperature

Item	Source Recall
Hydrogen	70%
Helium	28%
Carbon	1.5%
Nitrogen + Neon	0.5%
Red stars	Coolest $\approx 3000\text{-}4000\text{ K}$
Yellow stars	$\approx 5000\text{-}6000\text{ K}$; Sun yellow
Blue stars	Hottest $> 10,000\text{ K}$

TRICK - “H-He-C-NN = 70-28-1.5-0.5”

Source star-gas composition.

TRICK - “Red Cool - Yellow Medium - Blue Hot”

Colour-temperature order.

7. Black Hole + Chandrasekhar

- Black hole gravity itni strong ki light bhi escape nahi kar sakti
- Chandrasekhar limit = 1.44 solar masses
- Named after Subrahmanyan Chandrasekhar
- At/below limit: white dwarf can remain stable
- Sufficiently massive star beyond limit: after supernova collapse can produce neutron star or black hole

TRICK - “Chandra = 1.44 Sun”

White-dwarf stability limit.

EXAM TRAP -

Chandrasekhar limit 1.44 solar masses; nearest star fact se mix mat karna.

8. Constellations — U-O-C

- Constellation = recognisable group/pattern of stars
- Saptarishi = Ursa Major
- Orion = source notes guide toward Sirius
- Cassiopeia = important example

TRICK - “U-O-C = Ursa Major - Orion - Cassiopeia”

Saptarishi = Ursa Major.

9. Sun — 109 / 28 / 8:20 / Photo

Sun Fact	Source Value
Diameter	$\approx 109 \times$ Earth diameter
Gravity	$\approx 28 \times$ Earth gravity
Light travel time	8 min 20 sec
Visible surface	Photosphere
Energy source	Nuclear fusion: $H \rightarrow He$
Source composition	H 71%, He 26.5%, others 2.5%

TRICK - “Sun = 109 - 28 - 8:20 - PHOTO”

Diameter, gravity, light time, visible surface.

EXAM TRAP - “Photosphere Visible; Chromosphere + Corona Above”

Visible surface = Photosphere.

10. Planets Order — M-V-E-M-J-S-U-N

• Mercury → Venus → Earth → Mars → Jupiter → Saturn → Uranus → Neptune

TRICK - “M-V-E-M-J-S-U-N”

Sun se outward 8 major planets ka order.

EXAM TRAP -

Pluto = dwarf planet; eight major planets mein nahi.

11. Inner vs Outer — M-V-E-M / J-S-U-N

Feature	Inner / Terrestrial	Outer / Jovian
Members	Mercury, Venus, Earth, Mars	Jupiter, Saturn, Uranus, Neptune
Nature	Rocky, dense	Mostly gaseous / giant
Size	Generally smaller	Much larger
Moons/rings	Fewer	Many

TRICK - “Inner = MVEM Rocky; Outer = JSUN Giant”

Four-four planet split.

12. Planet High-Yield Facts

Planet	Source Memory Hook
Mercury	Closest + smallest; source notes no atmosphere; huge temperature variation
Venus	Hottest + brightest; greenhouse; retrograde rotation
Earth	Life + one moon + Blue Planet
Mars	Red due iron oxide; Phobos + Deimos; MOM
Jupiter	Largest; Great Red Spot; source lists 95 moons
Saturn	Least dense major planet; rings; source lists 146 moons
Uranus	Rotates nearly on side; tilt $\sim 98^\circ$; very cold
Neptune	Farthest major planet; strongest winds

TRICK - “Venus HOT; Jupiter BIG; Mercury SMALL”

Hottest =/closest.

TRICK - “Mars Red; Earth Blue; Uranus Sideways; Neptune Winds”

Colour + motion + weather hooks.

EXAM TRAP -

Hottest planet = Venus, not Mercury.

13. India & Space — Mission Chain

Mission / Satellite	Source Detail	Significance
MOM / Mangalyaan	Mars Orbiter Mission; 2013	India first interplanetary mission
GISAT-1	Geo Imaging Satellite-1	Source lists as India first satellite in geostationary orbit
APPLE	Ariane Passenger Payload Experiment	India first geostationary telecommunication satellite
Chandrayaan-1	Moon mission; 2008	Evidence of water molecules on Moon
Chandrayaan-3	Moon lander; 2023	Soft landing in lunar south-polar region; source notes first country near south pole
Aditya-L1	Solar mission; 2023	India first dedicated solar observatory mission

TRICK - “MOM = Mangalyaan = Mars = 2013”

India first interplanetary mission.

TRICK - “APPLE = Ariane Passenger Payload Experiment”

Full form ko exact yaad karo.

TRICK - “C1 Water; C3 South Pole; A-L1 Sun”

Chandrayaan-1 → water evidence; Chandrayaan-3 → south-polar landing; Aditya-L1 → solar observatory.

14. APPLE vs GISAT-1

Satellite	Source Distinction
APPLE	First geostationary telecommunication satellite named in source
GISAT-1	Listed separately as Geo Imaging Satellite-1 and India first satellite in geostationary orbit

EXAM TRAP - “APPLE = Telecom; GISAT = Geo Imaging”

Dono ko same label mat karna.

15. Rapid-Fire Revision

Question / Cue	Answer
Big Bang	George Lemaitre — 1930
Heliocentric	Copernicus — 1543
Geocentric	Ptolemy
Hubble Law	$V = H_d$
Age Universe	13.8 billion years
Our galaxy	Milky Way — Spiral
Nearest star after Sun	Proxima Centauri
Brightest night star	Sirius
Chandrasekhar limit	1.44 solar masses
Sun visible surface	Photosphere
Sunlight to Earth	8 min 20 sec
Hottest / largest / smallest	Venus / Jupiter / Mercury
Red / Blue planet	Mars / Earth
India first interplanetary	Mangalyaan / MOM
APPLE full form	Ariane Passenger Payload Experiment

16. Final Exam Traps — 12 Red Flags

- Big Bang = George Lemaitre; Hubble = expansion
- Red shift = away; Blue shift = approaching
- Milky Way = Spiral galaxy
- Nearest star after Sun = Proxima Centauri; brightest night star = Sirius
- Red stars coolest; Blue stars hottest
- Chandrasekhar limit = 1.44 solar masses
- Sun visible surface = Photosphere
- Sunlight reaches Earth in 8 min 20 sec
- Hottest = Venus; largest = Jupiter; smallest = Mercury
- Pluto = dwarf planet
- MOM = Mangalyaan = India first interplanetary mission
- APPLE = Ariane Passenger Payload Experiment; do not confuse with GISAT-1

EXAM TRAP - “Lemaitre/Hubble - Red/Blue - Milky - Proxima/Sirius - Star Colour - 1.44 - Photo - 8:20 - V/J/M - Pluto - MOM - APPLE”

Chapter ke 12 highest-yield traps isi sequence mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Universe 13.8 bya; Big Bang Lemaitre 1930, Hubble expansion 1925 + $V=Hd$, $H=67$; theories P-C-H-H-L = Ptolemy-Copernicus-Herschel-Hubble-Lemaitre → Red away, Blue toward; Andromeda blue-shift example → Galaxies E-S-I = Maffei-1 / Milky Way / Sextans A; Milky spiral arms blue, centre red → Stars: Proxima nearest after Sun, Sirius brightest night/Dog Star, Pole Star north; gases H-He-C-N = 70-28-1.5-0.5; Red cool, Yellow medium, Blue hot → Black hole light cannot escape; Chandra limit 1.44 Sun; constellations Ursa Major/Saptarishi-Orion-Cassiopeia → Sun 109×diameter, 28×gravity, light 8:20, Photosphere visible, fusion $H \rightarrow He$; source comp 71/26.5/2.5 → Planets M-V-E-M-J-S-U-N; inner MVEM rocky vs outer JSUN giant; Venus hottest, Jupiter largest, Mercury smallest; Mars red, Earth blue, Uranus side 98°, Neptune strongest winds; Pluto dwarf → India space: MOM=Mangalyaan 2013 first interplanetary; C1-2008 water evidence; C3-2023 south-polar soft landing; Aditya-L1-2023 solar; APPLE=Ariane Passenger Payload Experiment telecom vs GISAT-1 Geo Imaging.”

TOP 5: Lemaitre Big Bang vs Hubble expansion | Proxima nearest vs Sirius brightest | Sun Photosphere + 8:20 | Venus hot/Jupiter big/Mercury small | MOM + APPLE full form.